Transcript from National Grid Natural Gas Long-Term Capacity Report Public Meeting[1]

March 30, 2020

Coordinator: Welcome and thank you for standing by. At this time all participants are in a listen-only mode until the question-and-answer session of today's conference. At that time you may press star 1 on your phone to ask a question.

I would like to inform all parties that today's conference is being recorded. If you have any objections, you may disconnect at this time.

I will now turn the call over to John Bruckner. Sir, you may begin.

John Bruckner: Thank you, (Denise). And good evening, my name is John Bruckner, I'm the President of National Grid New York.

And I want to welcome you to our public meeting on National Grid's Long-Term Natural Gas Capacity Report, which was originally scheduled to be in person in Jamaica, Queens, fulfilling the provision of the company's November 2019 agreement with New York State. In that agreement, National Grid committed to researching and publishing a report that offers potential long-term solution options to address Downstate New York's recognized natural gas constrain issue.

[1] Provided by MyMeetings.

National Grid has issued this report in February of 2020 and scheduled a series of public meetings to hear from our customers and other stakeholders, and understand their expectations and opinions on how we together can solve the energy challenges that face our region.

Because of the outbreak of the coronavirus and in consideration of the wellbeing of the public and our employees, we've had to cancel the face-to-face public meetings, except for the in-person which was held on March 9th in Hicksville, Long Island.

To provide an alternative way to help you understand the report and our findings, we've scheduled a series of virtual meetings to achieve the same objective, to share information and answer any questions you have on each of the options, as well as hear from you directly through an oral comment period.

As with many companies and state agencies around the country, National Grid has activated its business continuity plans, and is in part leveraging teleconference and Webinar based technologies to stay connected with our stakeholders, communities, and employees.

And we've set up these virtual meetings to ensure we can still have a meaningful engagement with the public on these vital energy matters, and meet our obligations to New York State to have a long-term solution or solutions in place by the winter of 2021/'22, to meet the needs of the customers in New York City and Long Island.

Before we get to the run of show and how this particular session will work, I'd like to provide you with some context and other important information.

New York has seen dynamic economic growth in the Downstate region, expanding residential and non-residential building space and thousands of oil-to-gas conversions over the last 10 years. These factors have resulted in a substantial increase in the demand for natural gas, placing stress on our existing gas network and threatening National Grid's ability to meet our customers' needs when demand is at its peak.

This leaves little room for error in the face of unplanned supply interruptions or other contingencies. And as part of the November 2019 settlement agreement with New York State that lifted the moratorium on new gas connections imposed in May of 2019, we're taking measures to ensure we have enough supply for the winters of 2019/2020 and also the winter 2020/2021, including increasing our reliance on compressed natural gas, or otherwise known as CNG trucking, when needed to meet peak demand.

Beyond the next two winters, however, continued growth in demand for natural gas creates a challenge that must be addressed. The Long-Term Natural Gas Capacity Report presents 10 options to solve for that challenge.

It's important to note that the long-term solution options we're presenting, and the various combination of these options working together, have different implications for how long we continue to rely on CNG trucking to meet peak demand and whether we might have future restrictions on new gas connections.

Also of note, low-carbon solutions will be an important part of reaching longer-term clean energy goals. While formally not part of tonight's discussion, National Grid is partnering with third parties to develop renewable natural gas solutions, exploring hydrogen blending, and power-to-gas to help de-carbonize the gas system.

We're also piloting geothermal heat pump solutions with partners like NYSERDA. We continue to implement clean energy solutions throughout our entire service territory here in the U.S. with solar, wind, battery storage, and EV stations. All this can be found on our dedicated long-term capacity report Web site ngrid.com/longtermsolutions. In addition to the full detailed report, the Web site also features a shorter summary report to better help the public understand these options.

To further aid in transparency and understanding, we have updated the report on the Web site to include a technical appendix that lays out even more of the details of our analyses and assumptions, as well as estimates of how the costs of the different options would impact our customers over time.

The Web site also features other methods of sharing your feedback beyond tonight's public comment period. You can take a survey to share your views on various options. And I encourage you to do so. You'll also find instructions on multiple ways you can submit comments directly to the New York State Public Service Commission. Also on the Web site you can find information about how financial assistance programs for residential and small business customers who were inconvenienced because of the moratorium last year. Please visit this Web site for more information.

So let's begin tonight's meeting with some guidance on how the session will work. We'll begin the session by showing you videos on each of the options within each of the following three categories -- large infrastructure, distributed infrastructure, and non-infrastructure.

After we view the complete set of videos on the options, we'll open the phone lines for a 30-minute of Q-and-A session, specifically related to those options.

This is the Q-and-A portion of this meeting, your opportunity to ask any clarifying questions on any of the options. We have subject matter experts on the line to answer any specific question that you may have on any of the options.

I want to be clear. This is not the time to express your views and opinions. This will only be dedicated to asking questions that are clarifying questions on any of the options. When your line is open, we'll ask for your name, your ZIP code and affiliation, whether you're a customer or a part of an organization, and then you can ask your question.

In order to engage as many callers as possible during the 30-minute Q-and-A session, we will be limiting each caller to one question. And I understand some questions have various parts, and please feel free to do so, add those parts as well. The Q-and-A session will be recorded, transcribed and submitted unabridged to the New York State Public Service Commission.

And then finally, we will open the lines for you to have an opportunity to provide feedback and express your views during the public oral comment period. When you come on the line, we will ask again your name, ZIP code and affiliation. And then you can make your statement. We will be limiting all comments to two minutes. Your comments will be recorded, transcribed, and again submitted unabridged to the New York State Public Service Commission.

One final note, key to this meeting is a respectful exchange of information and ideas. While there may be some disagreement on how we get there, we are all here to solve for energy solutions that address the needs of current and future generations, and we look forward to hearing from you on how best to meet those challenges.

Okay. So now we'll screen the videos on each of the options. The videos will be played in the following order. Three large infrastructure options, which would include offshore LNG deepwater port; an LNG import terminal; and the northeast supply enhancement project. Then four, distributed infrastructure options, which will include peak liquefied natural gas facility; a liquefied natural gas barges; the Clove Lakes Transmission Loop Project; and the Iroquois Enhancement by Compression Project.

And then the last set of videos will include energy efficiency, demand response, and electrification. As you view and listen to the videos, please take the time to get your questions ready. Once we complete the viewing and listening to all the videos on each of the options, we'll begin the 30-minute Q-and-A session. You will need you to press star 1 to get in the queue to ask a question.

Thank you. And let's play the first set of videos.

Bill: We will now play the videos for the three large infrastructure options.

((Start of Video Presentation)) [All videos are available here: https://ngridlongtermsolutions.com/]

Man: Offshore LNG deepwater port involves the installation of an offshore buoy connected to one of the existing undersea pipelines that currently supply the Downstate New York region. Connected to this buoy would be a floating storage and regasification unit, an LNG bulk delivery ship.

This ship can store more than 3 million dekatherms of LNG and then vaporize it for injection into the pipeline, to accommodate peak demand and potentially help meet daily demands during the year.

Unlike a gas pipeline solution, this option stores a finite amount of natural gas and is replenished via ship-to-ship transfers. There are two potential locations for this to be installed. One, on the Long Island Sound and the other in the Atlantic Ocean off the Rockaways.

View shed impact should be minimal given the distance from the shore. Each location would be capable of delivering 400,000 dekatherms per day capacity. If this option comes to pass, it's highly reliable.

What would the impacts be on the environment? During construction, the impact includes disturbance of the seafloor, decreased water and air quality, increased sedimentation, noise and waste generation. Once operational, the floating storage and regasification unit and refueling tanker can be expected to generate stack and cooling water emissions.

Greenhouse gas, GHG emissions from an LNG system presents a climate impact 10% to 15% greater than a standard natural gas solution. Given its offshore location, the community impacts of this option should be minimal. We expect full implementation of this option within six to eight years.

If you read the report, you know each option carries some execution risks and considerations. Until this infrastructure comes online, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

And we have to implement a series of incremental energy efficiency, demand response and electrification measures to further reduce demand. If these measures do not predictably meet target volumes or timelines before the infrastructure comes online, there would have to be restrictions on new customer connections.

We estimate the total cost of this option, including the infrastructure and the demand side programs to be 1.9 billion in our low demand scenario and 2.22 billion in our high demand scenario.

Woman: A liquefied natural gas or LNG terminal involves building the infrastructure necessary to accommodate LNG carrier ships including an LNG receiving facility, onshore storage, regasification, and transportation components. Such a facility would be able to accommodate peak demand and help meet daily demand during the year. It would provide 400,000 dekatherms per day of natural gas capacity.

This is a highly reliable option. However, it is more vulnerable to severe weather events than the floating LNG and LNG barge options we've presented. This is an option all over the world and National Grid has experience with LNG facilities both here in the U.S. and in the U.K.

What would the impact on the environment be? It would be equivalent or slightly higher than those associated with the offshore LNG deepwater port, and higher than pipeline development. Greenhouse gas emissions would have a climate impact of 10% to 15% greater than a standard natural gas solution. There would be significant impact to the community, resulting from construction and operations.

And before it ever came to pass, it would require changes to or a waiver of existing New York State law that limits land storage of natural gas. Pending this necessary change to the law, we estimate the timeline for this option to be five to six years.

If you've read the report, you know each option carries some execution risks and considerations. Until this infrastructure comes online, we would have to retain our compressed natural gas trucking operations for periods of peak demands and we'd have to implement a series of incremental energy efficiency demand response and electrification measures to further reduce demand. If these measures do not predictably meet target volumes or timelines before the infrastructure comes online, there would have to be restrictions on new customer connections.

We estimate the total cost of this option, including the infrastructure and the demand side programs to be 2.46 billion in our low demand scenario and 2.78 billion in our high demand scenario.

Man: The Northeast Supply Enhancement or NESE project involves the construction of an interstate pipeline by Williams Companies, Incorporated to transport natural gas from Pennsylvania to New York through New Jersey via the Raritan Bay and Lower New York Bay. The pipeline would include approximately 23.5 miles, approximately 17 in New York, of underwater pipeline to the Rockaway Peninsula in Queens.

It will provide the Downstate region with 400,000 dekatherms per day of natural gas capacity. This option is extremely reliable and fundamentally resistant to above-ground weather events.

The environmental impact of construction would be similar to offshore LNG deepwater port, albeit over a larger geographic area. This impact includes disturbance of the seafloor, decreased water and air quality, increased sedimentation, noise, and waste generation. Ongoing operations will have a much lower effect on the environment and greenhouse gas emissions are expected to be 10% to 15% lower than any of our LNG options.

Community impacts would be minimal, with most of the construction happening offshore. There is some onshore construction in New Jersey, in brownfield sections in Williams' existing footprint. In the future, the carbon steel pipeline material would allow hydrogen to be transported, helping significantly assist with carbon reduction goals.

Assuming the option receives New York State and New Jersey permit approval by June 2020, we estimate this project can come online by December 2021. Because this has the fastest estimated infrastructure timeline, and once in place has very high reliability, it provides the highest degree of certainty that there will not be restrictions on new customer connections.

And once the infrastructure is online, we would be able to mothball compressed natural gas trucking and reduce some contracted peaking supplies, which would eliminate a source of supply with higher emissions and provide us with some contingency capacity.

This option would cost 1.82 billion in the low demand scenario and 1.83 billion in the high demand scenario.

((End of Video Presentation))

Bill: We will now play the videos for the four distributed infrastructure options.

((Start of Video Presentation))

Woman: A peak liquefied natural gas or LNG facility is one of our distributed infrastructure options. It would involve construction of a new LNG peak shaving facility plant and related infrastructure like what we already have in Greenpoint, Brooklyn and Holtsville, Long Island.

This is a highly reliable option and is in use throughout the United States. We have 12 of these facilities across our Massachusetts, Rhode Island and Downstate New York service areas. But to get this option up and running would require a change to or waiver of New York State law that limits land storage of natural gas. Once online, this option can potentially provide up to 100,000 dekatherms per day to meet periods of peak demand.

The environmental impact for this project would be moderate, including decreased air quality, pollution to storm water, and other runoff, disruption to natural resources and habitat, noise, and waste generation. It does, however, have a smaller footprint compared to our larger infrastructure LNG options.

This option would have to be sited within a community, most likely within an industrial zone. This option would have greenhouse gas emissions similar to our other LNG options and 10% to 15% higher than standard natural gas solutions. But remember, this option would only run to meet peak day demand.

Assuming changes or a waiver from current New York law, it's estimated a new site would be operational in five to six years. This option does carry some execution risks and considerations until it comes online. Because this solution only fills part of the gap between demand and supply and is estimated

to take five to six years to come online, it would need to be paired with a series of near-term and ongoing incremental energy efficiency, demand response, and electrification measures.

If these demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The estimated total cost of this option, including both the infrastructure and the demand side program is 1.49 billion in our low demand scenario and 2.54 billion in our high demand scenario.

Man: Liquefied natural gas or LNG barges are one of our four distributed infrastructure long-term options. This option involves the purchase and construction of one or more specialty LNG barges with vaporization equipment onboard. These barges would be towed to locations to allow them to transfer LNG from a variety of land and sea-based sources and inject vaporized gas into the existing onshore pipeline.

This option can provide potentially up to 50,000 dekatherms of peaking capacity per barge. If used in tandem, total supply would double to 100,000 dekatherms per day. This is a reliable option and the barges can be strategically placed in multiple locations, assuming permit approvals.

The environmental impact would be similar to the offshore LNG port solution but at a smaller scale, with disturbances of the seafloor, decreased water and air quality, increased sedimentation, noise, and waste generation. The only construction required is a new pier and short pipeline to an existing onshore

connection. In addition, there would be visual impacts to water views during the winter. And there may be loss of waterside recreation use in the immediate area due to the size of the barge system and security perimeter developed during the siting process.

We estimate the total implementation time for this project to be five to six years. Because this solution only fills part of the gap between demand and supply and is estimated to take five to six years to come online, it would need to be paired with a series of near-term and ongoing incremental energy efficiency, demand response, and electrification measures to reduce demand and completely close the gap.

If the demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The total estimated cost of this option, including both the infrastructure and the demand side program, is 1.36 billion in our low demand scenario, and 2.42 billion in our high demand scenario.

Woman: Clove Lakes Transmission Loop Project is one of our distributed infrastructure long-term options. It would involve construction of approximately 8 miles of new 30-inch field transmission names across the borough of Staten Island. It will help our ability to take more gas through the TETCO Goethals Take Station, and ultimately move more gas across our system to supply constrained areas.

When complete, it can provide approximately 70,000 to 100,000 dekatherms per day of baseload capacity to support daily demand. And it will be used to support the system year-round.

This is a historically very safe and reliable method of transporting natural gas. Most of the environmental impact will come from construction. It would involve installation of a new gas main across sections of a heavy populated urban environment and would possibly impact multiple highways and/or small water crossings during installations. Because of the proposed route, we would need a mitigation plan to minimize disruption to residents, businesses and traffic. Once operational, the impacts will be minor as little ongoing maintenance will be required.

Greenhouse gas emissions will be comparable to similar gas mains in the current distribution system. Once operational, this option will have lower emissions than the liquefied natural gas options or compressed natural gas trailers and trucking.

Initial estimates indicate that this option could be in service by November 2025 at the earliest, making the timeline five-plus years. Because this option only fills part of the gap between demand and supply and is estimated to take five-plus years to come online, it would need to be paired with a series of near-term and ongoing incremental energy efficiency demand response and electrification measures to reduce demand to completely close the gap.

If the demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The total estimated cost of this option, including both the infrastructure and the demand side program, is 1.57 billion in our low demand scenario, and 2.63 billion in our high demand scenario.

Man: The Iroquois Enhancement by Compression Project is one of our distributed infrastructure long-term options. It would involve the construction of additional compression facilities to increase capacity on the Iroquois Gas Transmission Systems, IGTS, existing infrastructure to transport natural gas from the U.S.-Canadian border at Waddington, New York through the existing system located in New York State, and Connecticut to the existing interconnection with our gas distribution system at Commack, Long Island.

When complete, this option can provide up to approximately 125,000 dekatherms per day of baseload capacity. We would split this capacity with Con Ed, so we would each have 62,500 dekatherms per day to our respective systems. This would enhance reliability to our system, especially to the gate delivery point serving our most eastern-bound customers. Above-ground weather events and manmade events such as traffic and automobile accidents would not impact availability of the natural gas supply.

This option will have a moderate impact on the environment. As construction will take place on an already existing line, the impact will be focused on locations already developed. As part of this project, Iroquois plans to further reduce methane and overall emissions at sites by installing low-nitrogen oxide and OX turbine units that will reduce NOx emissions by 40% of standard turbine units.

Potential greenhouse gas emissions will also be mitigated by Iroquois proposal to install methane recovery systems. All permitting, policy and

regulatory requirements must comply with FERC regulations and other appropriate federal, state and local agencies.

The project filed a formal application with the Federal Energy Regulatory Commission in February 2020 and is expecting a decision on approval by early 2021. Assuming timely approvals, the project could be in service by November 2023.

Because this solution only fills part of the gap between demand and supply and is estimated to take three years to come online, it would also be paired with a series of near-term and ongoing incremental energy efficiency demand response and electrification measures to reduce demand to completely close the gap.

If the demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The total estimated cost of this option, including both the infrastructure and the demand side programs, is 1.11 billion in our low demand scenario and 2.22 billion in our high demand scenario.

((End of Video Presentation))

Bill: We will now play the videos for the three non-infrastructure options.

((Start of Video Presentation))

Woman: Incremental energy efficiency is one of our non-infrastructure long-term options. It would involve building upon and expanding the base and incremental new efficiency New York programs with a more aggressive program offering that reduces annual energy consumption and design day demand. It will focus on gas efficiency savings, with a key focus on intensive weatherization measures, and will require extensive customer and trade ally engagement and training.

This option can potentially provide 111,000 to 216,000 dekatherms of design day savings. When implemented, this option will lead to passive energy and design day savings. It's key, however, that we're able to ramp up quickly and start realizing impacts by winter of 2021/'22.

There is no additional environmental impact as work will only be completed on existing structures. Overall, this could result in a reduction of greenhouse gas emissions from less natural gas consumption.

To be successful, we'll need to further develop a wide range of contractors and suppliers who will need to hire additional employees to support the spending over the duration of the program. These investments will go directly into the Downstate economy and any build savings will go directly to customers.

This option will require a significant increase in level of effort across the entire energy efficiency network, including growth in number of contractors, staff in local permitting offices, and increases in company employees. The program would have to scale up by 2025 to become similar in size to our current successful program in Massachusetts.

This solution requires a series of incremental energy efficiency, demand response and electrification measures that start to have predictable impact on

peak demand by the winter of 2021/'22, and continue to grow their impact at a size and pace that enables existing National Grid gas supply to meet customer needs.

If the demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The total estimate cost for this option is 1.51 billion in our low demand scenario and 2.62 billion in our high demand scenario. This includes the total cost for the entire no-infrastructure portfolio of energy efficiency, demand response, and electrification required to close the gap between supply and demand.

Man: Demand Response is one of our non-infrastructure long-term options. Compared to other non-infrastructure options, Demand Response is more cost-effective and can be ramped up more quickly. That's because Demand Response will leverage existing equipment at customer premises.

The overall challenge with this option is related to the overall number of eligible customers we have, and the challenges that multifamily and commercial customers have in maintaining a backup source of heat, typically fuel oil, for use in temperature control or non-firm programs.

Those who fall under the temperature control category include those multifamily and some commercial and industrial customers who we ask to switch off of natural gas and move to another fuel source, typically fuel oil,

when the temperature drops below a certain level. For example, 16 degrees Fahrenheit.

We also have thermostat setback programs where we provide an incentive for residential customers to turn down their thermostats during cold weather events. Implementing this option would require new rate design programs that encourage these customers to remain on non-firm rate programs.

If implemented, we estimate approximately 81,000 to 108,000 dekatherms of design day savings.

One of the risks with this option is that there is no guarantee that customers will switch to their alternate fuel as they can still use natural gas. Accordingly, the number of participants during an event can fluctuate and not meet projections, thereby putting reliability at risk. We feel, however, this could improve over time as the Demand Response program matures. There is little or no incremental impact to the environment with this option.

In addition, there will be a limited impact to the climate as Demand Response will reduce energy use during peak periods, but some of this displaced energy will swing to non-peak periods. Meanwhile, non-firm customers switching to an alternate fuel, typically fuel oil, will result in higher greenhouse gas emissions. In the future, fuel oil could be transitioned to renewable fuels such as biofuels, which would help reduce environmental impact.

This option requires New York Public Service Commission approval, including incentives and total investments, before the program can begin. And it will impact permitting policy and regulatory activities at the local and state level. This solution requires a series of incremental energy efficiency, demand response, and electrification measures that start to have predictable

impact on peak demand by the winter of 2021/'22, and continue to grow their impact at a size and pace that enables existing National Grid gas supply to meet customer needs.

If the demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The total estimates cost for this option is 1.51 billion in our low demand scenario and 2.62 billion in our high demand scenario. This includes the total cost for the entire no-infrastructure portfolio of energy efficiency, demand response, and electrification required to close the gap between supply and demand.

Woman: Electrification is one of our long-term non-infrastructure options. This option presents us with an opportunity to reduce design day natural gas consumption by converting customers' base heating energy source from natural gas to electricity. We could achieve this by using cold climate heat pumps, which operate efficiently even at cold temperatures and they can be installed and operated without a fossil fuel backup heating system.

If we assume a robust incentive program is in place and effective incremental electrification can yield a range of 52,000 to 86,000 dekatherms per day in design day demand reductions. To achieve these reduction levels, we estimate we'll need 5000 to 15,000 incremental customers per year, starting in 2021, to move to electric heat cooking and industrial use.

Once implemented, this is a reliable option to reduce forecasted design day demand. It's important to know that if customers retain gas for other uses, however, such as for cooking, water heating, or as a backup heating source, the design day savings will be less than anticipated.

There's minimal current environmental impact with this option. However, the reduction of gas consumption will be largely offset by an increase in electricity consumption, with some studies showing higher current emissions for electric power than natural gas in the short term. Longer term, due to the forecasted de-carbonization of New York electric supply, incremental electrification will result in decreased emissions.

For electrification to really take hold, we'll need to create an ecosystem that will include a wide range of contractors and suppliers who will need to hire additional employees to support the spending over duration of program. And it will mean growth in total electric customers and electric demand, requiring permitting, policy and regulatory requirements to be in place, equating to significant startup costs and resources.

The biggest challenge with electrification right now is both the upfront and ongoing costs. The current cost of installation is often twice as high as the typical natural gas heating unit. In addition, electric prices are relatively high in Downstate New York and would likely lead to overall increase in customers' annual energy utility bill. For this option to be economic in this region, we'd have to see big increases in heat pump efficiency or big reductions in electric prices.

Electrification requires a series of incremental energy efficiency, demand response, and electrification measures that start to have predictable impact on peak demand by the winter of 2021/'22 and continue to grow their impact at a

size and pace that enables existing National Grid's natural gas supply to meet customer needs.

Note, if the demand reduction measures do not predictably meet target volumes or timelines, there would have to be restrictions on new customer connections. In addition, unless demand reduction can predictably exceed targets, we would have to retain our compressed natural gas trucking operations for periods of peak demand.

The total estimate cost for this option of 1.51 billion in our low demand scenario and 2.62 billion in our high demand scenario. This includes the total cost for the entire no-infrastructure portfolio of energy efficiency, demand response, and electrification required to close the gap between supply and demand.

((End of Video Presentation))

Bill: John, the videos on each of the options is now completed.

John Bruckner: All right, thank you, Bill. We will now begin the Q-and-A session, which will last 30 minutes, and be followed by the public comment period. Again, we have subject matter experts on the line to answer any specific question you may have on any of the options cited in the report.

I just want to be clear; this is not the time to express your views and opinion. We will follow up this section with the public comment period. This section will be dedicated to asking clarifying questions on any of the options. When your line is open, we will ask for your name, ZIP code and affiliation, and then you can ask your question.

So at this time, operator, can we get the first question?

Coordinator: Thank you. We will now begin the question-and-answer session. It is star 1. Please unmute your phone and record your name clearly in order to be introduced with your question. One moment please.

John Bruckner: Thank you.

Coordinator: And up first is (Michael Grossman). Please state your affiliation and ZIP code.

(Michael Grossman): Yes. (Michael Grossman), 07840 (unintelligible) and electric customer at 13436.

John Bruckner: Go ahead, (Michael). Yes, (Michael). Your question is?

(Michael Grossman): Yes. My question is, after all of the different scenarios presented, wouldn't it be of your opinion that the least environmentally obtrusive methodology and most reliable would be the Northeast Pipeline?

John Bruckner: Hi, (Michael). I guess I'd put that into two categories. Let me answer it this way, and then, again, I'd welcome you to join because it sounds more like a public comment issue that you're raising and an opinion, so. But as far as, you know, which one is the least environmental or the most beneficial, I think it's premature right now to say what that decision is. So we're providing all the facts on all of the options.

Part of this process is to get the public engagement in this and be part of the solution going forward, for not only our customers and not only our current customers but also for future generations. So I think it's a good question. I

think it's a little premature for us now to say the best option is this. We're going to fulfill this process, get public opinion, and synthesize all that with a set of recommendations to the Public Service Commission in the State of New York. But thank you for your question.

Coordinator: And up next is (Tanya Jones). Please state your affiliation and your ZIP code.

(Tanya Jones): Sure. My name is (Tanya Jones) from HeartShare Human Services of New York. My ZIP code is 11421.

My question is, do you have any idea what the prospective impact on water quality is with the NESE pipeline?

John Bruckner: All right, (Tanya). So, as I mentioned before, we have a number of subject matter experts, and for that, I'm going to refer that to Robert. Robert, are you on the line?

Robert: Yes, I am, John. Yes, thank you for your question. The NESE project has been FERC-approved, and as you know, it's (unintelligible) for the water permit with New York State and New Jersey DEC that has been denied. Williams is the company doing the construction of this proposed project. And obviously there are concerns with water quality issues, and they are dealing with the two state entities to work through those concerns currently. So that's where it stands.

John Bruckner: So, Robert, I think it would be safe to say that, until the water quality issues are addressed with respect to those permits, you know, Williams would not have those permits from the DEC in New York or New Jersey. So, receipt of those permits would then therefore suggest that, if they did get them, that the

water quality issues were addressed successfully. Is that a fair statement? Thank you.

Okay. Operator?

Coordinator: Up next is Alexi Assmus. Your line is open, and please state your affiliation and ZIP code.

Alexi Assmus: My name is Alexi Assmus, Eastern Environmental Law Center, 08542.

With the assistance of (Synapse) Energy Economics, the Eastern Environmental Law Center submitted questions to National Grid asking for specifics of how various assumptions to estimate future demand were developed and how they all fit together to produce the demand forecast. When should we be expecting responsive answers to these questions from National Grid?

John Bruckner: Okay, Alexi, thank you for your question. I just want to make sure that I have it clear. I think that the theme of the question was the calculation of future demand and how that was determined, and the assumptions that were built into. And I know you referenced the (Syntex) Energy Economics, but again I think the question was more directed towards how was the - what are the assumptions and how is the demand going forward for natural gas determined?

So if I could, I'd like to have Cameron address that from the report standpoint. And certainly, Pete, if you have anything you'd like to add, please do. But, Cameron?

Cameron: Sure. Thanks, John, and thanks for the question. So in the Long-Term Capacity Report, we lay out the assumptions that we used with regards to showing some of the historic demand and some of the key drivers of that demand looking forward at some of those same key drivers and how they may change over time in terms of their magnitude, and thus, the volume growth, how that may change over time.

And then we also look to take into account the impact of things like additional energy efficiency, demand response, and electrification, to arrive at a demand forecast that has essentially two scenarios that kind of creates a low demand and a high demand forecast.

To the extent that there have been comments regarding our assumptions and how we created that demand forecast, we are in the process of, through the course of these virtual public meetings, also through the course of accepting written feedback, we are getting all of those comments. We are accepting comments through April 17th.

And once that comment period is complete, we then intend to publish an update, any update that we might make, the long-term capacity report, as well as a response to comments where we will take, you know, the categories, the themes of the comment, and respond to those comments.

So the expectation is that we're, you know, looking, after we receive all those comments in April 17th, that sometime thereafter we would be publishing that response and the updated report. So, probably by the end of April is sort of our target.

John Bruckner: Thank you, Cameron. And Pete, again, if you had anything you wanted to add or not.

Pete: Well, I would add to this that the estimates of our growth are based on historic growth that we've seen as well. Historic growth from the system has been about 2.4% per year.

And notwithstanding the additional efforts that we're making to increase energy efficiency and demand response programs, that growth would be anticipated to continue at about a 1.8% growth per year. The energy efficiency that we've been applying would reduce that to about 1.1% per year, or in the highly successful case, we would be down to about 0.8% per year.

But this growth is being driven by high levels of oil-to-gas conversions and takes into account all of the significant construction that is taking place in the Downstate area. We've added a significant amount of square footage in the areas such as Downtown Brooklyn. Due to all of this construction, the demand for gas is high. It is an affordable choice for a lot of developers. They are opting to go for natural gas as their fuel of choice. And as such, the demand for natural gas continues to be significant.

John Bruckner: Thank you, Pete. Operator?

Coordinator: Up next is (Barbara). Please state your affiliation and ZIP code.

(Barbara): Hi. My name is (Barbara). I live in Brooklyn and my ZIP code is 11211. And I'm a rate payer.

And my question is, is National Grid really serious about, if their tunnels don't get built, just to go for the non-invasive way of, you know, changing late people get gas and, you know, lowering their gas usage, which is a fossil fuel?

John Bruckner: So, (Barbara), I want to make sure I understand your - your question is really about, and operator, if we can keep (Barbara) on the line, I just want to make sure I understand the question that, is National Grid really focused on, I think you're referring to the non-infrastructure options?

(Barbara): Yes.

John Bruckner: Okay. So let me have Owen and Keith talk about some of those non-infrastructure options and what the opportunity is that can be derived from them, and whether or not they in of themselves could actually solve for the, you know, long time as a whole. So, Owen, can I ask you to address (Barbara's) question?

Owen: Absolutely, John. So, within the non-infrastructure solutions, there are three different options -- demand response, energy efficiency, and electrification. And we are already aggressive pursuing all three of those as part of the natural evolution of our business and as a path to optimize the way that we utilize the gas system today to meet as many customers' energy needs as possible.

And I've already filed for incremental energy efficiency programs and expanded demand response portfolios within the rate case that were submitted to the Public Service Commission in April of 2019.

So we are committed to these as tools that we think we can use to achieve the best possible outcomes. They do come, as was outlined in the videos, with pros and cons. All of these options come with pros and cons. And we are trying to figure out which set of solutions or which suite of solutions will provide the best outcome for all customers (to have) determined through this public process.

Keith, did you have anything you wanted to add from the energy efficiency side?

Keith: You covered it very well. Thank you, Owen. Just, you know, the energy efficiency, we have done - we've succeeded very nicely in (unintelligible) the efficiency programs versus the targets that we've set with the Public Service Commission. And those targets will continue to grow.

This set of incremental energy efficiency, in terms of weatherization, in terms of installation and (unintelligible) will be even above those higher targets. So we are dedicated to moving things in this direction. And as Owen said, it's with a suite of solutions that will get us to the outcome that we're looking for.

John Bruckner: Yes. (Barbara), the only thing that I think is important is I want to eke out of your question the word serious, right? So, is National Grid serious?

So, you know, let's - I think it's important that we recognize where National Grid is with respect to energy efficiency and demand side management programs. We are consistently year on year considered among the top leaders in this space nationally.

So, while Owen and Keith talked about targets that we have with our public service commission, with is our regulator, and meeting those targets, above and beyond that, National Grid continues to be a leader in the energy efficiency space, delivering far greater than our targets, but more importantly, greater than our peers in the utility space across the nation.

As a matter of fact, I believe among all National Grid companies in the U.S., in New York, Massachusetts and Rhode Island, we are clearly in the top 10 in all of those operating companies nationally. So, yes, definitively, to answer

your question, we are serious and we do believe and are committed and continue to be committed to energy efficiency and demand side management.

Also, our demand side management or our demand response programs on the gas side of the business are not only, you know, top in the nation, they are groundbreaking. We clearly have taken the initiative far greater than any of our peer utilities as far as demand response programs. So I hope that answers your concern as well as your question.

Operator?

Coordinator: Up next is (Margaux). Please state your affiliation and ZIP code.

(Margaux): (Margaux), 11222. You're a private company and you need to maximize profits to compensate your shareholders. How different would the scenario (unintelligible) look if profit wasn't part of the equation? For instance, you're applying to the Public Service Commission to increase our rates by $185 million to complete the North Brooklyn Pipeline.

That gas is for sale to other states, not for North Brooklyn. If you didn't need to make a profit and you just needed to provide power, would that pipeline still be necessary?

John Bruckner: So, (Margaux), I think that's a good question, and really with respect to profit and the North Brooklyn Pipeline, the MRI project. The MRI project is being constructed to serve the residents of Downstate. It's not being constructed to serve any other regions or any other state. It is to supply the needs and the demand that's Downstate. So that's number one.

With respect to National Grid shareholders and all of our stakeholders, our Public Service Commission holds us to a standard of delivering the most cost-effective and safe energy for our customers. And so, is there a return for our shareholders? Yes, there is a fair, reasonable return. That's determined by our regulator.

And as far as the cost to our customers, that too is determined by our regulator, to ensure that we do, at the business of serving the customers in the most efficient, cost-effective and publicly safe manner. So I believe that answers your concern.

Pete: John, if I may add one thing to your response. I'd like to…

John Bruckner: Sure, Pete.

Pete: …clear that the MRI pipe that is being referred to in the question is not a supply project and does not add any supply to the Downstate region or any other region. It's possible that the person asking the question may have heard something different than that. But the MRI project is not listed amongst these options here because it does not actually add supply to the Downstate region. It only makes the system operate more efficiently. So I'd like to clear that up for anyone that may be listening, that MRI is not a supply project.

John Bruckner: Thank you, Pete. And I believe that was (Margaux) that asked the question, and again, just for clarity, the benefit of the MRI project is for the reliability of the Downstate region, which is the Brooklyn area as well. So, thank you, Pete, for that clarification.

Coordinator: Up next is (Will Lange). Please state your affiliation and your ZIP code.

(Will Lange): Hi, this is (Will Lange) with Water Furnace, ZIP code 46809. Was really interested in the videos. Two quick questions.

First of all, the impact of the gas supply was all measured in dekatherms. The impact of electrification and the last several options was measured in - it wasn't really measured.

Secondly, is there any way to view the calculus of how it was determined that electrification would create higher electric - excuse me - higher total energy bills for Downstate customers? Is that directly from the TRM?

John Bruckner: Yes. So, well, just two things there. One is, how is the electrification measured versus gas in therms? And also, how is it determined as far as the costly - the cost aspect or ramifications of electrification.

So I do have Kenzi on as a subject matter expert. Kenzi, can you address those two questions for (Will) please?

Kenzi: Yes. Thank you. Thank you for those questions. So, regarding the question about the assumptions made about electrification, we used some assumptions about the cost of the system as well as assumptions about energy costs that are common in Downstate New York, to develop sort of an overall cost of heating for customers that would participate in electrification.

I am not certain whether or not those numbers can specifically from the TRM, numbers surrounding efficiency. But we benchmark those numbers against what's sort of common in the marketplace today.

And then regarding the dekatherms versus other units, dekatherms are standard measurement common in use for the natural gas industry. They are comparable to MMBTUs almost a one-for-one comparison.

Thanks for your question.

John Bruckner: Thank you, Kenzi. Cameron, did you have anything you wanted to add in the report?

Cameron: No. I mean, I think Kenzi did a great job covering it. I think that, you know, in the report we've tried to lay out the assumptions that go behind all of our - looking at the different options, the associated costs. And we do take the - each of the different options down to what we think it can achieve in terms of an MBTA (through) thousands of dekatherms level, to try to make a comparable on an apples to apples basis.

I would say that we also have put online more detail in what we call a technical appendix. So, our technical appendix, which is available, also gives a little bit more detail behind our assumptions. And so if people are interested, they can access that to be able to see more details behind what went into the report.

John Bruckner: All right, good. Thank you for your question, (Will). Operator, next question?

Coordinator: Up next is (Lee Ziesche). Please state your affiliation and your ZIP Code.

(Lee Ziesche): Hi. (Lee Ziesche). I'm a rate payer. I'm with the Sane Energy Project, and it's 11216. I have three questions. Should I ask them all at once or is it better to separate them?

John Bruckner: You can ask them at once, I'll see if I got them all. And if I do, we'll proceed with answering them, (Lee).

(Lee Ziesche): Cool, sounds good. So my first question is, were the impacts of extraction of the gas considered in any of your environmental assessments of the proposed projects, particularly like the impacts of fracking on water and air quality in Pennsylvania, as well as upstream methane leakage?

My second question is, there was a 20-month investigative report that came out in Rolling Stone this year that found that companies like (Exxon) have known for decades that radioactivity is building up in gas infrastructure (at scale), particularly in pipelines, and things that have pumps like compressor station.

Since more of National Grid's gas is coming from the Marcellus Shale which has the highest level of radioactivity in the country, is National Grid doing anything to ensure that radioactivity isn't building up in any of the proposed gas infrastructure and your current distribution system?

And then my third question is, in the estimated costs that have been presented to us in this report, is the rate of return for National Grid included in those costs? And if so, what is that proposed rate?

John Bruckner: All right. So I think I got them all, (Lee). Your first question was the impact of extraction on things such as the water and other systems, the upstream impact of extraction. The radioactivity proposed from your Rolling Stone article that you referenced. And also the estimated costs or the rate of return included, is it included in the cost estimates.

So I'd ask Cameron to address the cost issue, and as well some of the other issues I'd ask Pete if he has any insights as to some of those concerns as well upstream off of National Grid's system, upstream on the supplier side. So, Cameron?

Cameron: Yes. So, when we estimated the cost of the solution, you know, we are taking into account what we believe the total cost of the solution will be. I cannot speak specifically to any exact rate of return when it comes to National Grid. I mean, that would have to be something that would be dealt with in any sort of rate case and with the regulators.

But we are looking at an estimate that is the, you know, the expected cost impact, so that we can make sure that we are comparing across the different options around the total cost as well as the, you know, making it sort of apples to apples, from a comparison perspective.

John Bruckner: And as far as any of the impacts, I don't know if the study talks about that, any of the upstream impacts on where the gas is actually coming from. Cameron, is that included in the study?

Cameron: We did not assess related to cost of - insofar as, excuse me, environmental impact of extraction. We did not do any sort of environmental study on extraction. We have looked at comparing the environmental impact of natural gas relative to oil, relative to LNG, relative to compressed natural gas, liquefied natural gas.

And we've also showed some comparisons of how that might change over time as you could introduce things like renewable natural gas or hydrogen and how that also compares to both the marginal emissions as well as average emissions over time with regards to electric and electric heat.

John Bruckner: Okay. Thank you, Cameron.

Kenzi: And John, if I might jump in on talking about lifecycle methane emission calculations, as a company, we've done and we're doing a lot of work in trying to accurately evaluate lifecycle emissions for natural gas, as well as working with the entire supply chain to address those emissions. And so, if you're interested in learning more, I suggest that you look at a collaboration called One Future.

It's in partnership with the EPA. And it's working with producers, it's working with transmission companies, and it's working with distribution companies, to come up with best practices for how to reduce emissions, calculate emissions, and then report those as accurately as we can with the tools that we have year after year. That's something that National Grid participates in and we're a founding member of.

We're also member of the Natural Gas Supply Collaborative. And I recommend you look into them as well if you're interested in learning more about what upstream producers can do to address emissions. The Natural Gas Supply Collaborative is the group of large natural gas customers who have come up with ESG, environmental, societal and government, measures that we recommend producers use to evaluate their production methods. So those are a few things the company is doing to work with our upstream partners in addressing the total lifecycle impact of emissions from natural gas.

John Bruckner: Terrific. Thank you, Kenzi.

Coordinator: And up next is (Sarah). Please state your affiliation and your ZIP code.

(Sarah): Hello. My name is (Sarah), I'm (unintelligible). My ZIP code is 11237. I was just wondering if any of the 10 plans incorporate renewable energy and/or solar power. And if not, do any of them have the potential to integrate renewable energy and/or solar power?

John Bruckner: So, (Sarah), it's a good question. A couple of things when you talk about renewable energy. So we talked about it upfront, and I'm going to have Kenzi address this as well, and certainly open up to any of our other subject matter experts.

But when you talk about renewable energy solutions and the manner in which we serve customers, we will be getting and continue to lead in the renewable energy space. So I know I spoke about it on some of the other prior calls, and I'll mention it here, that National Grid is in partnership with New York State, with the Newtown Creek wastewater treatment facility. That facility is New York City's largest wastewater treatment facility.

And we have a collaborative effort with them to eke out the renewable natural gas that's a byproduct of their process, and put that back into the natural gas system. So it is considered renewable natural gas.

In the space of solar, we continue to partner with a number of organizations on solar projects here Downstate as well as Upstate. Downstate we've partnered with a third party at National Grid to serve (LIPA's) request for proposals that they had put out on the market, and we're currently in the process of developing a solar farm.

As far as battery storage, again, serving (LIPA's) needs Downstate here, National Grid has partnered again with a number of partners in the industry

and have successfully put online two large battery storage facilities, both on the east end of Long Island, to address Long Island's electric concerns as well.

Off of Long Island, we continue to be a very major player in offshore wind, being the company, the first company in the United States to successfully connect an offshore wind project to the grid. And we have a battery storage facility up in Massachusetts service territory that's among the largest in the U.S., and it's currently online, so, offsetting the need for additional infrastructure and so forth.

So we are doing a lot in the space of clean and renewable energy, and we continue to do so. Certainly Downstate here. We are a partner and the only utility representing - with representation on the offshore wind consortium, which is, as you probably are aware of, Governor Cuomo has clearly taken a leadership role in offshore wind development here in New York State, and National Grid is a founding member and a committee member on the National Offshore Wind Farm Consortium to help the development of offshore wind. So, a lot of places where National Grid is taking leadership positions in renewable energies, and will continue to do so.

But as I said at the start, many of those, while clean energy, renewable energy, and certainly a part of the energy plan going forward in the portfolio of meeting the region's energy needs, not one of the issues right now on the table to address the long-term gas capacity constraints. Some of it is, and I'm going to defer to Kenzi to talk to some of those issues. Kenzi?

Kenzi: Yes. Thank you, John, and thank you for the question. So I think to start in answering that question, it's really important to clarify what the demand represented in this long-term capacity report, is it's really demand tied to our heating needs in New York State.

You know, it's people heating their homes in the winter, it's businesses heating their buildings, it's industrial organizations using fuel for their processes. It's, you know, cooking and running your dryer. It's being focused on heating, rather than plug loads like charging cellphones or charging electric vehicles.

And so when we're talking about what renewables can address this issue, sometimes it's hard to find the renewables that overlap with heating. And John spoke to Newtown Creek where we're looking to integrate renewable natural gas, which is something that can be directed towards heating because it's compatible with existing appliances, it's compatible with our existing natural gas infrastructure, and can go to meet that demand.

We're also looking and working on developing more solutions to develop similar low-carbon fuels. So, for instance, limiting that sort of brings together traditional renewables like solar and wind. And this issue of renewables for heating is a process called power-to-gas. It's a process where you take excess electricity. You run it through a system called moleculizer and you use it to break water into hydrogen and oxygen. This is a process that we're really critically evaluating.

We actually, we had proposal to demonstrate it in our Downstate rate case, because we think it can take and beneficially use the renewables that the state is working so hard to invest in, like offshore wind, take excess production, we don't use year-round because load doesn't always match - demand doesn't always match supply, and be able to redirect that towards our heating needs in the state.

So there are some innovative things that the company is working on to begin to bring, in addition to your traditional renewable natural gas from places like

wastewater treatment plants, more renewables into the natural gas network and to help bring down the total sort of carbon impact of that system.

Thanks for your question.

John Bruckner: Thank you, Kenzi.

Coordinator: And up next, we have (Charles Ryan). Please state your affiliation and your ZIP code.

(Charles Ryan): (Charles Ryan), 07756. For the CNG trucking fleet options, what is the fuel source for the trucks and what are the emissions per truck if you have it per day or for year?

John Bruckner: Sure, Charles. I'm going to ask Pete to give us some information on that, and Cameron, if you have anything you could add, that'd be great. But, Pete, can you answer that question?

Pete: So, typically the trucks themselves are a standard diesel tractor that is - and the fuel source for that would be your normal diesel outpost. Your - the actual gas that is inside of the truck can be coming from one of several stations in Pennsylvania, New Jersey or Upstate New York.

The gas that is on a CNG trailer must come from upstream of our gas system in order to be incremental to the daily deliveries. So, those trucks do need to bring gas from some distance away into our service territory in order to be effective in the purpose that they need to meet.

John Bruckner: Thank you, Pete. Cameron, you have anything you want to add?

Cameron: Just with respect to CNG, I mean it's something that, you know, we have used in other places, and it's now something that we've started to use in Downstate New York to make sure that we have enough gas during periods of peak demand. It's been in place for the last I think three winters.

And it is something that, as we've seen growth and constraints in our network where we have had to add CNG capacity, and expect to continue to have to add CNG capacity, at least through next winter, to make sure that we have enough supply on those periods of peak demand.

John Bruckner: Thank you, Cameron.

Okay. Go ahead. Go ahead, operator.

Coordinator: We do have the last question from (Ed). Please state your affiliation and your ZIP code.

(Ed): My ZIP code is 11697, and as a rate payer, that's my affiliation. My question is, it seems to me that the ultimate goal here of National Grid is to put this pipeline in. I understand you come up with a bunch of solutions. A few of them have already been knocked. But here's the question.

Since the Marcellus Shale where this fracked gas will be coming from happens to be the most radioactive, higher than the Gulf Coast, higher than the Permian formation in Texas, and even higher than the Bakken in North Dakota, what are you going to do to safeguard the health of people who live surrounding the compressor station since the (radon) is going to be - it's going to be there. So, what can you do or what is the plan to protect those people? Thank you.

John Bruckner: All right. (Ed), thank you for your question. I am going to ask Robert to give us some insight as to your question with respect to safeguarding the public around the compressor stations. But your opening question or statement was about the solution and the solutions identified. And some of them may have been tried in the past.

And I think it's an important point. National Grid is not selecting any particular solution at this point. What we're doing is we're trying to obtain all public comment, stakeholder engagement, and get everybody's opinion, and everybody's recommendations as to the direction that we should go to solve the region's energy challenge. And so, you know, I wouldn't agree that we've decided on a solution. This is part of the process to ascertain the best solution.

And with respect to some of the options may have been tried in the past, you're absolutely right. (Ed), some of these options may have been tried in the past. But we want to make sure that it's an inclusive process to look at all the available options.

And so, some may require, you know, when we talk about the risk of execution, some may require additional permitting or change of law as it's constructed today. But again, we want it to be fully inclusive. And we'll deal with those challenges with respect to execution risk. But we want to have them all out on the table so that we all have transparency and visibility to all the options.

With that, that's the reason why we want to make sure that we include options that may have been tried in the past, but maybe recognizing -- and it has been recognized -- the need for incremental supply to the region. Maybe that dynamic will change, and maybe a solution that may have been tried in the past may find its way to be a solution tomorrow.

But with that, let me ask (NESE) or Cameron if you have anything you want to add with respect to how Williams or what Williams' plans are, how they're addressing their compressor stations.

Robert: Sure, John. With respect to Williams' role in the supply that would be entering our system, Williams is the transporter of gas, so they're not the producer of the gas, of the fracked gas that the caller was talking about.

And regarding their compression stations, definitely Williams has been proactive in reducing emissions at stations that are increasing in capacity. They do this by installing electric compression, in some cases, to reduce noise, and other emissions, and also increasing efficiency at those compressor stations.

I can't speak to the studies some of the callers are referring to regarding radioactive contamination or (radon) and things like that. But Williams is, you know, fully permitted on the FERC side for this project, and they're working with the states on the water permit.

And National Grid is not really involved in those direct discussions that are (unintelligible) so we can't speak to some of those details that the caller was pointing out. Like (Transco) is the supplier, they're the transporter of the gas, they're not producer of the (gas).

John Bruckner: All right. Well, (Robert), thank you. And (Ed), the other thing I would just mention is that, you know, you're referencing some of the solutions, and I guess the theme there is that it's a pipeline solution. There are many options here that we've discussed that some are pipelines, some are non-pipeline solutions.

So, you know, obviously the compression station may apply, in one case to a pipeline solution, but not necessarily to, you know, whether it's an LNG solution or a non-pipe solution. So there are many options here. And I just want you to be rest assured that we are looking at all of the options. And we really and truly believe, as one of the earlier callers asked, are we serious?

Yes, we are serious about non-pipeline solutions. Our track record has demonstrated that and we'll continue to demonstrate our leadership role in energy efficiency and demand response as part of a bigger, broader solution.

So with that, operator, I think that concludes the Q-and-A portion of the call.

So at this point we're going to move to the public comment period. And I just want to just lay out that we'll begin the public comment period now, and it allows you to present a public statement on the options presented in the report. We'll have a listening panel which will include Cameron. Cameron was the author of the long-term report. Melanie Littlejohn is an executive at National Grid here for our New York State Customer and Community Management organization.

And tonight, Melanie Littlejohn will be facilitating the public comment period. So I'd like to hand it over to Melanie. Melanie?

Melanie Littlejohn: Good, thank you. Thank you, John. And…

John Bruckner: You're welcome. You're welcome.

Melanie Littlejohn: …good evening all.

Please note that during this section, the public comment period is now open and is only limited to statements. We will not be taking questions. And therefore, the Q-and-A portion of the program has concluded. As a reminder, to get in queue, please dial star 1. Once the operator opens your line and you come on the line, please state your name, ZIP code, and affiliation for the record. And then certainly you can make your statement.

We will be limiting all comments to two minutes. Each comment will be recorded, transcribed, and submitted unabridged to the New York State PSC. And if you believe you need more time to expand on your statement or if you have further questions, we encourage you to visit ngrid.com/longtermsolutions to upload a written statement or any other question directly to the New York State Public Service Commission.

And as a reminder, please dial that star 1 to make your comments. Operator, first caller please.

Coordinator: That comes from (Billii Roberti). Your line is open. Please state your affiliation and ZIP code.

(Billii Roberti): My ZIP code is 11746, my affiliation is Mothers Out Front and the New York Geothermal Energy Organization.

My comment is, of all the eight options considered by National Grid, seven are fossil fuel projects, that will be built with gas utility customer money. New York Climate Law requires 40% reduction of greenhouse gases by 2030 and the elimination of greenhouse gases by 2050. How can National Grid propose to bring more methane into New York? I say no to truck and train transportation of LNG. They are not nicknamed bombtracks and bombtrains for nothing. They are dangerous as well as costly.

I say no to more compressor stations for the Iroquois pipeline. We do not need more fracked gas. We need less. Your report shows that non-fossil fuel options can meet the growing demand for thermal energy and contribute to the phase-out of fuel oil (within) National Grid's (services). It is possible and it must be done.

The people of New York have spoken. We do not want the NESE pipeline. We want non-pipeline solutions. For National Grid to stay viable in a fossil fuel reducing world, it needs to seriously consider a new business model that includes geothermal district heating and cooling using heat pumps. A tiny demonstration project was conducted in Riverhead. What were the results? I never heard them.

I'm surprised you mentioned cold climate air source heat pumps. These can cut National Grid (unintelligible) heating market. That is quite the opposite with ground sourcing pumps. By installing and owning geothermal district heating and cooling loop systems, these systems will provide income for the utility income both winter and summer into the foreseeable future. If this option is currently prohibited by law, we can help the utilities change the law.

The concept is this, gas lines are already in place, the right-of-way use by the utility can be repurposed and gas lines removed as geothermal loops are installed. This is a much better investment because it will be building a more sustainable business model instead of simply maintaining leaky gas lines and other infrastructure that have limited future. (Unintelligible) assets benefit no one.

Building new pipelines and gas infrastructure is the way of the past. A business model that provides heating and cooling to business (via a fee) to

access geothermal heat - loops, geothermal loops, is the way to keep National Grid alive and well into the future when fossil fuels are just a footnote in the history books.

Your bankruptcy will serve no one. Your continuation as a new business model is a win-win. Thank you.

Melanie Littlejohn: Thank you, (Billii). Operator, next caller?

Coordinator: That is from (Michael Grossman). Your line is open. Please state your affiliation and your ZIP code.

(Miguel Grossman): Yes. (Michael Grossman), 07840, and a rate payer of electrical heat pump use in 13436.

And as a heat pump user, I can tell you that electrical use is still very high for heating systems. And the only reasonable solution that I could comprehend would be the Northeast pipeline that would disturb the least amount of environmental problems and be an adequate and reliable supply to all of New York City for the oncoming future. Thank you very much.

Melanie Littlejohn: Thank you, (Michael). Operator, next caller?

Coordinator: That is from (Tanya Jones). Your line is open. Please state your affiliation and ZIP code.

(Tanya Jones): Sure. My name is (Tanya Jones) from HeartShare Human Services of New York. My ZIP code is 11421.

HeartShare has partnered with National Grid for over 20 years in helping the community in those most vulnerable. We have found National Grid to have a huge community conscience. If there is minimal impact on water quality, NESE seems like the best choice here out of all the choices - out of all the choices proposed. New York cannot afford for customers not to have access to heat or gas in any situation, and low-income customers can't afford additional electricity prices. Thank you.

Melanie Littlejohn: Thank you, (Tanya). Operator, next caller please.

Coordinator: That is (Jackie Weisberg). Your line is open. Please state your affiliation and ZIP code.

(Jackie Weisberg): Affiliation is 350 Brooklyn and my ZIP code is 11201.

Going to read something that I was just noticing on - published at Vox. Gas utilities are proposing to increase their renewable natural gas (with) new pipelines and to charge their customers for the (unintelligible) rate hike. In a relatively short span of time (unintelligible) electric movements have emerged as dozens of (unintelligible) to encourage all-electric construction in new buildings. Natural gas utilities do not like this movement one bit.

The more all-electric buildings there are, the fewer natural gas rate payers there are. An all-electric future inevitably involves the obsolescence or at least a substantial diminution of natural gas utility. Naturally they are fighting back (seriously) with so-called (unintelligible) groups, PR campaigns and lobbying at the local level.

Their main argument has to do with renewable natural gas, an industry term for methane, captured from biogenic wastes at landfills, livestock operations,

farms, sewage treatment. Utilities argue that ramping up the production of RNG and (blanking) it with natural normal - normal natural gas and pipelines can reduce greenhouse gases faster and cheaper than electrifying buildings.

According to the Southern California Gas Company spokesman, a growing number of experts in industrialized nations recognize the need to include renewable gas. By pursuing electrification (unintelligible) regulators are pushing on this very rate hike.

It will be nice for you to believe that this is true, but it's not. RNG is not (unintelligible) the industry claims. Its local air and water impacts are concentrated in vulnerable communities. And even if it were low carbon, there isn't enough of it to substitute for more than a small fraction of natural gas. And even if it were low-carbon equitable and abundant, it still wouldn't be an excuse to expand natural gas infrastructure or slow electrification.

It isn't a close call. The research is clear, especially in temperate climates like California, RNG is not a viable alternative (unintelligible) de-carbonizing buildings. It is a desperate bid by natural gas utilities to delay their inevitable decline. Policymakers would be foolish to fall for it.

Melanie Littlejohn: Thank you, (Jackie). Operator, next caller?

Coordinator: That is from (Barbara). Your line is open. Please state your affiliation and ZIP code.

(Barbara): My name is (Barbara), it's 11211. And I'm a rate payer. And I just want to rebuff some of the people who said that the NESE pipeline is wonderful, the most environmentally sound thing, when it's really one of the worst ones in the books.

It, you know, goes to the seafloor, which, you know, these things are better left underneath the ground and not into the water. The boats also sound like a nightmare for Rockaway. And the beaches will get polluted by the, you know, there's benzene and all these other chemicals that are - that have been sitting there for a long time in the seafloor. So that's what the pipeline will do.

And then there's birds that nest on Staten Island and (unintelligible) on the Jersey Shore. They don't like construction, so they're gone, and stuff like that.

The other thing is that these meetings are not transparent. They are - it's very limited. It has to have people that have computers. And since the COVID, you know, 19 is here, you can't have your friends over to "Let's do this meeting together and listen on a phone." You know, it's like, you can't do that, unless we're all six feet apart (unintelligible) to have meetings with people, and my friends don't even want to go out.

So, I mean, it's really bad. I think you really should cancel your next meeting and have some meetings in May and just have your report come out in July or August. These projects are all like four, five years, six years down the road. So, why do you need to have all this stuff together? This is what I don't understand. So, please, you know, cancel your meetings and, you know, face the people. Thank you. Bye.

Melanie Littlejohn: Thank you, (Barbara). Next caller please?

Coordinator: That is from (Kimmy Chong). Your line is open. Please state your affiliation and your ZIP code.

(Kimmy Chong): My name is (Kimmy Chong), my ZIP code is 10002. And I'm the (unintelligible) in the past two decades, I do a lot of job for (unintelligible) and my (unintelligible) oil because the gas you provide as a clean (energy) and also is (unintelligible). If you ask the people using the electrical heat, it will cost them a lot. So that's why I strong support the National Grid to do more gas work for us. And we need gas. We need a lot of gas in the future. Thank you very much.

Melanie Littlejohn: Thank you, (Kimmy). Next caller please.

Coordinator: And that is from (David Vassar). Your line is open. Please state your affiliation and ZIP code.

(David Vassar): Hi, I'm (David Vassar). I'm volunteer with Food & Water Watch, 10027 is my home ZIP.

So, looking at the long term, we really, really need renewable energy in place, much sooner than later. New York State Climate Law now requires net zero greenhouse gas emissions by 2050. And Local Law 97 will reduce building emissions 40% by 2030. That's the projection.

Moreover, gas use will actually plummet during that time, so this pipeline, or barge, or LNG station, whatever the infrastructure is, will actually become obsolete well before its intended retirement date. So we'll have a lot of infrastructure in place basically just rusting and polluting just by its presence and, you know, with no longer with a job to do. So it's a very short-term plan in terms of profitability anyway.

National Grid's proposed new fracked gas delivery infrastructure may possibly feature some enhancement to safety and environmental impact mitigation, but

investment in fracked gas infrastructure of any kind, whether it's pipelines, LNG stations, or power plants, runs totally counter to the spirit and the mandate of the Climate Leadership and Community Protection Act signed by Governor Cuomo last year. And it's - that act is our best hope right now to secure a future that is sustainable for our kids and for generations to come.

So I urge you, please either put your operating capital, your engineering talent, which is considerable, and your manpower, into creating renewable energy, or please get out of the energy delivery business altogether. Thank you.

Melanie Littlejohn: Thank you, (David). Next caller?

Coordinator: That is from (Monica). Your line is open. Please state your affiliation and ZIP code.

(Monica): Hi, my name is (Monica), I'm from Sane Energy Project. And my ZIP code is 11213.

So we are in this historic moment of a global crisis when every single individual, industry and government official should be doing every single thing possible that we can to prioritize keeping each other safe. And right now you are doing the opposite.

You are pushing forward quickly and irresponsibly making these hearings inaccessible while people are focusing on survival, literally on surviving, and putting us all further at risk while this unnecessary, dangerous pipeline pushing through North Brooklyn and even more fracked gas projects that go against New York Climate Law, the Williams NESE pipeline has already been

halted twice because of public opposition, and we will stop it again if you continue to try to push it forward.

So that we can see that you're focusing on profit instead of community needs. And this is a terrifying time that we're living through. There's so much unknown that it is abundantly clear that this is not the time to double-down on fossil fuel infrastructure, especially when we're in the middle of a congested, overpopulated city that is the epicenter of the COVID-19 crisis.

We know that fracked gas pipelines explode and the infrastructure can have terrible health impacts, especially respiratory, and we can see the tremendous burden already in hospitals and medical workers that are quickly running out of space and resources, setting up hospitals in Central Park. If there's another future health emergency we can prevent, we have to do it.

This is a really scary time that isn't a time to get smarter and to get better and to restructure our thinking and unbind ourselves from energy systems that are outdated, harmful, and wasteful. This is a time for us to wake up and make every effort we can to move us towards renewable energy that is truly for people and by people. Thank you.

Melanie Littlejohn: Thank you, (Monica). Operator, next caller please.

Coordinator: That is from (Joe Barone). Your line is open. Please state your affiliation and ZIP code.

(Joe Barone): (Joe Barone) Shale Directories, 19072. I support NESE and I'm going to take an article that appeared on (Axios) Web site today. And it says the top American Civil Rights Activists are opposing an abrupt move away from

natural gas, putting them at odds with environmentalists and progressive Democrats who want to ban fracking.

In recent interviews, Reverends Al Sharpton, Jesse Jackson, and National Urban League President Marc Morial said energy costs are hitting people of color unfairly hard. These concerns expressed before the coronavirus pandemic are poised to expand as paychecks shrink across America (unintelligible) I think people are concerned about affordability and they are concerned about being left in the cold, Sharpton said in early March.

I think natural gas is a temporary. I don't think we ought to make it an end-all and be-all solution, but in the interim people in communities of color should not pay the brunt of suffering through the cold winter.

I support the call to ban fracking with proper transition, Jackson said in February. In the meantime, those who are down and out have to have natural gas in the meantime. Fracking, the controversial oil and gas extraction method, has unlocked vast reserves of natural gas across the country over the past decade.

That's providing a cheap, relatively cleaner fuel for electricity, heating and manufacturing. Lower income people spend a disproportionate amount of their paychecks on heating, electricity and transportation. This is especially so in Black and Hispanic communities where poverty rates are higher.

In looking at where the country is today with the pandemic and the tremendous demand on people's wallets, I think the NESE pipeline is the best solution that will be the bridge to get to the renewable resources. Thank you.

Melanie Littlejohn: Thank you. Thank you. Next caller please.

Coordinator: That is Alexi Assmus. Your line is open. Please state your affiliation and ZIP code.

Alexi Assmus: Yes. My name is Alexi Assmus, Eastern Environmental Law Center, 08542. And I'd like to make a comment regarding the need for the NESE pipeline.

Preliminary findings from a report by a team of energy experts at nationally regarded (Synapse) Energy Economics, a report that will be made public within the next several weeks, show that there is no need. National Grid estimated gap between forecast demand and available supply is inappropriately calculated.

(Synapse's) business-as-usual analysis, using more appropriate fuel conversion rates, taking into account NYSERDA's current energy efficiency program, and showing that natural gas capacity is greater than National Grid has estimated, finds no gap between forecast demand and supply in Long Island, and no need to build a pipeline.

Another (Synapse) preliminary finding is that National Grid's no-infrastructure solution is the least cost option, if NYSERDA's energy efficiency programs are properly taken into account.

I'd also like to just make one comment. I did ask a question in the question-and-answer period. We sent in draft discovery questions to National Grid, and they were very specific. For instance, we asked, please provide the details behind the construction of the baseline demand for cash shown in figure one, and we wanted - we asked, please provide the actual changes in load each year associated with each of the five customer types listed in Table 9, etcetera. We had a list of seven questions to that specificity.

And the response I got in Q-and-A was that the comments would be held and there would be some general responses. But in the Q-and-A, it didn't seem that the people on the phone understood the question that we had. We have very specific data-driven questions that will inform our final independent analysis of need.

Melanie Littlejohn: Thank you, Alexi. Operator, next caller.

Coordinator: That is from (Marina). Your line is open. And please state your affiliation and ZIP code.

(Marina): Thank you. This is (Marina), ZIP code 73754.

One of the solutions proposed to supply the energy need is trucking in compressed natural gas to New York City. There's significant risks involved in transporting CNG over the road and major impacts on traffic of adding dozens of large trucks to the streets of New York City daily.

New York City has historically relied on heating oil delivered by trucks to residential areas, which caused significant traffic congestion. One of the benefits of the NESE natural gas pipeline is practically zero impact on traffic. And NESE also offers unsurpassed reliability. In a situation like the present lockdown, traffic is interrupted, which would potentially leave New York residents without critical gas supply as they rely on CNG trucks. NESE, on the other hand, would continue to provide affordable, safe and reliable gas supply to New York residents.

Solar and wind power, and other non-fossil fuel projects are important for the future, but they cannot supply the immediate needs of New York City. The

process of solar panel manufacturing, wind turbine manufacturing, and battery manufacturing also have environmental impacts, and the manufacturing processes rely on fossil fuels. Ultimately, solar and wind are not nearly as reliable sources of energy as NESE natural gas.

NESE is the most reliable, effective solution for New York today and in the future. Thank you.

Melanie Littlejohn: Thank you, (Marina). Operator, next caller please.

Coordinator: That is from (David). Your line is open. Please state your affiliation and zip code.

(David Gronim): Yes, (David Gronim), 11803, rate payer. Yes, I have been living and traveling in many states in the U.S. and provinces in Canada, so I see many people are using, have been using natural gas as a renewable, reliable gas. When I move here in Long Island back in 2017, I was extremely disappointed. I really felt I was living in back in 1950s because I see there are a lot of people in Long Island that are still using (unintelligible) oil for heating their homes. I cannot even go out to my backyard and enjoy my time because of the (unintelligible) gases coming from my own (unintelligible) neighbor.

I would certainly support the NESE project (than) any other renewable gas. Thank you.

Melanie Littlejohn: Thank you, (David). Operator, next caller please.

Coordinator: That is from (Drew Poling). Your line is open. Please state your affiliation and ZIP code.

(Drew Poling): Hi. This is (Drew Poling), I'm from 11226. I'm calling to urge the (unintelligible) of the Northeast Supply Enhancement Project and any other fossil fuel related projects.

Obviously, a big part of this planning session is because of the faulty moratorium that National Grid imposed on New York citizens, really just continue to show that they don't actually have regard for citizens' health and safety in mind when they're proposing these projects, made clear by seven of the eight options being fossil fuel related, and especially the fact that NESE has already been blocked numerous times because of the impact of the locality.

So it's pretty clear that we need to stop proposing more fossil fuel-based solutions and commit to finding a renewable alternative. Governor Cuomo has also made it clear at the CLPPA that New York is going to be a climate leader, it's just a question of if National Grid can be part of that transition or not. Thank you.

Melanie Littlejohn: Thank you, (Drew). Operator, next caller?

Coordinator: That comes from (Elizabeth Halliday). Your line is open. Please state your affiliation and ZIP code.

(Elizabeth Halliday): Thank you. My name is (Elizabeth Halliday), Clean Capital Leadership Council, 11930.

I support the NESE proposal. Natural gas will prove the most reliable and low-cost, low-emission energy source as we retire fuel infrastructure that emits more greenhouse gases. I also support natural gas being used in concern

with renewable energy sources, most especially offshore wind as we're located in the zone for the (south) wind farm.

Natural gas will be the lower emission alternative for the next 20 or so years. And if used, is the reason for lower emissions currently enjoyed in the USA. We seek to continue this good progress and the country will lean more on natural gas while renewable energy infrastructure is developed and deployed.

If COVID-19 alerts us of anything, it is that regulation and response needs to be more nimble and rapid, allowing for the fastest deployment of the best solutions. My group also supports all of the above approach to clean energy infrastructure, but natural gas is the most mature and reliable technology to meet regional demand. Thank you.

Melanie Littlejohn: Thank you, (Elizabeth). Operator, next caller?

Coordinator: At this time I have no one waiting. It is star 1 if you would like to have your line open for your statement. And please record your name in order to be introduced. Once again, star 1 for your line to be open for a statement. Thank you.

Melanie Littlejohn: While we're waiting for our next caller, we want to just remind everyone this evening is, whether you attend a virtual event or not, that there will be an equal opportunity to have your voice and your opinions heard and reflected prior to any final decisions being made. And you can do so by accessing our Web site at ngrid.com/longtermsolutions. And there you can review the supporting materials and submit a survey or provide feedback to the New York State Public Service Commission.

So, again, please we encourage you to go and review the full report, the summary, the videos, and you'll also see information regarding the full summary of the report of all the meetings and the number of attendees and the summary of every comment unabridged, and pertinent detail that all of you have provided.

So again, we encourage you to go visit ngrid.com/longtermsolutions, to again add any additional feedback or view all of our material.

Operator, are there any callers?

Coordinator: Yes. I do have two. (Eileen Berry), your line is open. Please state your affiliation and ZIP code.

(Eileen Berry): Hi, I'm a citizen, 11554. I've been a resident of New York State for 54 years. The house that I grew up in had natural gas. It's wonderful, very clean, very safe, no problems.

And so I got married and bought my next home, it is reliant on oil and propane to cook. I have been in contact with everyone for the last 17 years to attempt to get natural gas brought to my street, hooked up to my house, to protect our neighborhood from these trucks that are carrying in all these oils. We've had spills in the neighborhood. We've had such environmental problems. They're worried about natural gas pipeline with environmental problems, look at these trucks that are delivering oil to all the neighborhood. It's hurting our neighborhood.

You have to let this non-NESE option, just get rid of it. Let National Grid go ahead and supply us with the natural gas we need here as residents. Really, I'm very disappointed with all this moratorium and having to delay and delay

and delay. Seventeen years I'm waiting for my home to be hooked up, and I'm using all this other junk that you're worried about is going to be environmental hurting. I'm hurting the environment in my home because you'll not let me use natural gas. Thank you.

Melanie Littlejohn: Thank you, (Eileen). Next caller please.

Coordinator: That comes from (Isabel). Your line is open. Please state your affiliation and your ZIP code.

(Isabel), your line is open.

(Isabel): Yes, hi. How are you doing? I'm a pay rate citizen. And I support the NESE pipeline. That's it. Thank you.

Melanie Littlejohn: Thank you, (Isabel). Operator, next caller please.

Coordinator: That comes from (Joe Ludwig). Your line is open. Please state your affiliation and your ZIP code.

(Joe Ludwig): I'm a local plumber, 11703. As far as the pipeline, I think that would be the best way to go. They drill under the sand, they put tunnels in, and everything else. If they start doing this - the solar things or the wind towers out in the ocean, they're going to be turning up to beds out there, which I say is a cleaner energy, but I think that will make more of a mess of our oceans.

As far as people with the oil heating, they can't afford geothermal, so when their boiler goes, instead of converting over to natural gas, they're putting in - the oil back in, which is the cheapest way to go. They have more pollution from that than anything. That's pretty much it. I think natural gas is the way

to go for Long Island. It's been here for a long time, and hopefully it will be here for a lot longer.

Melanie Littlejohn: Thank you, (Joe). Operator, next caller please?

Coordinator: That is (Laura Shindell). Your line is open. Please state your affiliation and your ZIP code.

(Laura Shindell): Hi there. (Laura Shindell), ZIP code 10019. I'm from Food and Water Action.

Just to respond to a few of the - the misinformation that other callers have been privied (sic) to and called about, natural gas is a really nice PR term for fracked gas. It's not environmentally friendly. The process of extracting it has been causing water to be poisoned, earthquakes that happen in regions that are - never had earthquakes before.

And the primary component of natural fracked gas is methane, which is as much as 84 times more effective at trapping heat in the atmosphere than carbon dioxide over a 20-year period. So it is not more environmentally friendly.

And for the folks that have been saying, oh, well, we've had it forever, about, you know, eight, nine years ago, the United States started moving to fracking as the primary method to extract gas, so, yes, you may have had gas in your homes years ago, but the method of extraction is much more dangerous. And now we are also in a climate crisis, thanks to the years and years of fossil fuel usage.

New York State has already weighed the pros and cons of, you know, continuing with fossil fuels or moving to renewable energy, and elected officials in the state and the citizens of the state overwhelmingly support moving to renewable energy because of all of the reasons I just cited.

So I'm sorry to hear that some callers have taken the propaganda bait from the natural gas industry, who by the way have paid the same sort of PR professionals that the tobacco industry used to convince people that tobacco was safe, and we know how that all worked out. But the NESE pipeline and all of the fossil fuel proposals in the report are terrible options.

And it's not a choice between natural gas and oil. We need renewable energy. The point isn't to stay on oil; the point is to move to renewables. And we have to stop NESE pipeline and National Grid has to get onboard or be put out of business. Because the whole reason we're even having these meetings is because Governor Cuomo threatened to revoke your license to operate because of the lies you were spreading in New York State. Thank you.

Melanie Littlejohn: Thank you, (Laura). Operator, next caller?

Coordinator: That is from (Simon Staley). Your line is open. Please state your affiliation and ZIP code.

(Simon Staley): My name is (Simon Staley) and I'm a resident of (NYCHA). And I'm just hoping that the gas lines do get through. They give us a chance to use natural gas. Thank you. And my ZIP code is 11238.

Melanie Littlejohn: Thank you, (Simon). Operator, next caller please.

Coordinator: That is from (Ed). Your line is open. Please state your affiliation and your ZIP code.

(Ed): Yes, hello. Rate payer, 11697.

I'd like to voice my opposition for the NESE pipeline. It will be especially harmful to my community in the Rockaways. From the carcinogens that will be released from the seafloor to contaminate our waters and seashore, to the damage to wildlife that frequent these waters. It only gets worse.

The real problem is not the pipeline, but the damage that an increase in fracked gas will cause. We were proposing pulling up a tremendous amount of organic matter that is now trapped deep within the earth, to burn, and exacerbate climate change. None of these proposals are a solution. What kind of planet are we going to leave our children?

Scientists tell us fracked gas is 86 times worse as a greenhouse gas than CO2. The planet is warming, creating more powerful storms. I witnessed firsthand as a first responder the death and destruction caused by Super Storm Sandy. My town Breezy Point had 132 hours burnt to the ground. I personally watched people die. This is something you never forget. National Grid should not be a partner in the destruction of our planet. Thank you.

Melanie Littlejohn: Thank you, (Ed). Operator, next caller please?

Coordinator: I have no one in queue. And as a reminder, it is star 1 if you would like to present a statement. Once again, star 1.

Melanie Littlejohn: Okay, while we are waiting for additional callers, again, as a reminder, for those on the call who do not have access to the Internet, please be aware that

you can still send your feedback by U.S. mail. The address and the person that you need to contact is the Honorable Michelle L. Phillips, Secretary to the New York State Public Service Commission, at 3 Empire State Plaza, Albany, New York, 12223-1350.

Again you will send it to the Honorable Michelle L. Phillips, Secretary to the New York State Public Service Commission, 3 Empire State Plaza, Albany, New York, 12223-1350.

And as a reminder, after the public meetings are completed and all the (unintelligible) is collected and reviewed, the company will issue a supplementary report that summarizes and includes public and customer input, all to enable an agreed long-term solution with New York State by June 2020. And this timeline will enable the solution or solutions to be in place and in operations by winter of '21/'22.

Operator, do we have any additional callers?

Coordinator: We do. Next is (Herb Mens). Your line is open. Please state your affiliation and ZIP code.

(Herb Mens): Hi. I'm just another customer. I'm watching the Williams pipeline and other hydro fracked gas, poisonous, destructive construction destroying our earth and our future, and result in the destruction of our birds and our bees and our trees around here in Staten Island. I've lived here for many, many decades, and it's just getting worse and worse and worse.

The gas that's being used is obviously irradiated, so that means that we are not only using an irradiated gas but also heating our homes. And then this hydro-fracked waste is being dumped in our aquifers, and we are now breathing and

drinking and bathing and planting in irradiated, toxic waste. The toxins that are used have been outlawed in 17 countries, on every continent, except for here, because of the kickbacks and the payoffs and the criminals in government and the sociopaths who are running this company, National Grid.

We need to break this company apart. We need to arrest and prosecute the people who are running it. And we need to make this a 100% renewable state with the next few months. It can be done easily. The governor and the - and the government in general can simply give each household enough money to build a - or install a, you know, a, what you call those cells, you know, the solar cells. And we could be off fracked gas forever, which is what we should be doing, instead of poisoning and destroying our children's lives and futures and poisoning our own system here.

Melanie Littlejohn: Thank you, (Herb). Thank you.

Coordinator: And up next is (Cathryn). Your line is open. Please state your affiliation and your ZIP code.

(Cathryn): Hello. I'm a rate payer. I live in Greenpoint, 11222. I am a very concerned citizen. I am terribly aware that we are in the midst of a climate emergency, we're in the midst of a pandemic. What we need to do is mobilize all forces to confront the future and work together to create genuine sustainability in our state and in our nation and in our globe. And it can begin in New York State.

First of all, I'm upset that, you know, people are under the impression that there's a choice. We have the NESE pipeline or we can have more oil. That's not genuine. There's many options that you presented, most of them were not about renewable, only the last option offered any kind of renewable options. But that is where we should be focused.

There were no geothermal - there's not a geothermal option there. You're talking about raising rates of customers and to the tune of getting $185 million to finish phase five of the North Brooklyn MRI pipeline. That money could be used instead to invest in something like geothermal infrastructure.

There was a man who called and he said, well, people won't have geothermal because it's too expensive, so they get oil. That doesn't seem like a real choice that people are confronted. May I have geothermal or can I have oil? People aren't even being offered that.

And so your - National Grid presentation keeps talking about solutions and I'm not seeing like real solutions. We don't have time. We don't have time to get - to wait till 2050 to get to net zero. Time is running out. We are in a climate emergency. Climate crisis is threatening. World population, it's, you know, and our children's future is at stake. The planet's future is at stake. And National Grid, if you want to be a part of the future and the solution, you have to completely change these paradigms that you're setting out and offer many more infrastructure and investment solutions that are non-invasive.

Melanie Littlejohn: Thank you, (Cathryn).

(Cathryn): Thank you.

Melanie Littlejohn: Thank you.

Coordinator: And I have no other callers at this time. As a reminder, it is star 1 if you would like to have your line open.

Melanie Littlejohn: As a reminder, while we wait for additional callers to join us, we know that many of our residential and small business customers were inconvenienced because of being denied natural gas service last year.

We have a customer assistance program for those who experienced financial hardships to help. So if you submitted a completed application for gas service between May 15, 2019 and October 11, 2019, you may be eligible for bill credit and other financial assistance, and we encourage you to review our programs by visiting nationalgridus.com/gasconnect-assist.

Coordinator: Thank you. And I do have another caller, that is (Laura Shandel). Your line is open, please state your affiliation and ZIP code.

(Laura Shindell): (Laura Shindell), Food and Water Action, ZIP code 10019.

I would just like to note that the financial assistance program that I think Melanie was just mentioning for people impacted by the moratorium, just for the record, I believe that money is coming from the mandated penalties that National Grid was forced to pay by the state because of how (royally) they screwed up with the moratorium and how they misled their customers and used their customers as pawns.

So it's not out of the goodness of National Grid's heart that they're providing national - providing rate payers - I'm sorry - providing financial assistance to impacted people, but rather because they have been legally obligated in order to maintain their contract and license to operate. Thank you.

Melanie Littlejohn: Thank you, (Laura).

Coordinator: And I do have another caller, (Herb Mens). Your line is open. Please state your affiliation and ZIP code.

(Herb Mens): I'm just a client of yours, a customer of yours, 10301, here in Staten Island.

I am dismayed, profoundly dismayed that National Grid is having these kinds of meetings in the middle of a crisis, an international COVID-19 crisis. And obviously these meetings should have been postponed until the public and elected officials had the bandwidth to, you know, to engage in these processes.

And this is a public health issue, National Grid is creating a public health issue on top of another public health issue. They're bringing in bombtrains and drilling pipes all over Brooklyn and everywhere else, in our oceans, in our bay here. And now they want to put another one in - here in Staten Island.

This is a horrible situation. The pipelines should not exist. The Williams Company should be shut down. And these people should be prosecuted. They should be arrested and locked up.

Basically we're dealing with a group of criminals who run National Grid and the Williams Company and etcetera, who are establishing a - taking monies to establish a genocidal project. These projects are, gas fracking projects, are deadly for people.

They create a situation that destroys life on the planet, forever. You can't make this, you know, back again. Once you've destroyed the marine life, the fragile marine life, and pumped in toxic mercury and arsenic and lead and PCBs, and dug up everything, it's over with.

I mean, the lowest half-life on this is 25,000 years.

Melanie Littlejohn: Thank you…

(Herb Mens): …group of people who are…

Coordinator: Thank you. And up next is (Brenda). Please state your affiliation and your ZIP code.

(Brenda): Oh, hi. Can you hear me. I'm (Brenda), citizen, 10706. Can you hear me?

Melanie Littlejohn: Yes, we can hear you, (Brenda).

(Brenda): Okay. I'd like to pick up on a couple of things that the last caller was saying and (Barbara) was saying very early on in the comments and the first caller (Billy). The procedure on this is all wrong. We should not be having any of these meetings. These meetings should be halted until this pandemic is over. This is completely unfair. Forget democratic, it's just not even in the realm of fairness. People need to be able to attend these meetings and to know what is going on regarding these pipelines and what they're really about.

The other thing I wanted to - so the procedure is all wrong. It's all wrong, okay? These meetings should not be happening. There should be a halt, a complete moratorium on this topic, until this pandemic is over, and we can really get together to discuss this.

Then, the other thing that needs to be addressed is the information versus really the misinformation that is being - that National Grid is not addressing. And clearly, many of the callers are misinformed. Natural gas is not a renewable energy. It is not renewable at all. And neither is fracked gas. And

this is fracked gas. Oil peaked in 2006, natural gas peaked in 2010. We are now going towards fracked gas because there is no getting extracting natural gas in any other way except to frack it. Okay?

And (Eileen) needs to understand that it's not a matter of oil versus gas. Geothermal and heat pumps are an option and they are not being weighed in (unintelligible) in the National Grid proposals. These are absolutely good options for Manhattan is built on bedrock. Geothermal, according to two scientists who I've discussed with, that is the perfect location to use geothermal.

It is not a question of, you know, natural gas is a great idea for now or it's a good bridge or fracked gas is a great option. It's not an option. It's not an option. We're in the middle of a pandemic and this is directly related to environmental issues. And anyone who does not admit that is only lying to themselves.

Melanie Littlejohn: Thank you, (Brenda). Thank you.

Coordinator: And up next is (Barbara). Your line is open. Please state your affiliation and your ZIP code.

(Barbara): Hi. I'm a rate payer and my ZIP code is 11211. And I want to respond to a couple of people who think all these fracked gas is just dangerous and it's horrible and people don't think about the end-product where children are getting weird diseases in Pennsylvania and it's very disturbing what's happening in these drilled areas by the Williams Company and every other gas company that runs, you know, goes into these places.

The other thing is I have no gas products in my house and I have an induction stove and I really like it. It's really great. It's wonderful. I have electric heat and it's wonderful too. And it's all working really well. I do not want any natural gas in any of my property ever again. And that's that.

And these meetings are a sham, and I hope that you see to it to have real meetings with real people that can really talk straight. Thank you.

Melanie Littlejohn: Thank you, (Barb).

Coordinator: Thank you. And there are no other callers at this time. It is star 1 if you would like to have your line open.

Melanie Littlejohn: Okay. We'll just hold and wait for the next caller.

Coordinator: I do have a caller. (Herb Mens), your line is open. Please state your affiliation and your ZIP code.

(Herb Mens): Okay, thank you. I'm a customer, 10301.

To continue the dialogue here, pick up on what other people have spoken about, all of National Grid's fracked gas proposals should be turned down. These proposals run the gamut of offshore liquefied natural gas terminals, onshore liquefied natural gas. I mean, this is - this stuff is like rolling in the atomic bomb and just destroying everything for 100 miles, 20 to 100 miles around you, including the deaths of children and women and men.

When things go off or they leak, they have deadly consequences, rare cancers, all kinds of horrific problems, Parkinson's and dementia for elders, birth

defects. The people in Pennsylvania are suffering terribly. Their land values have gone down to nothing.

And the people who are behind this must be prosecuted. They must be prosecuted and they must pay with their own money. We must have a - the management and the investors and the people, the politicians, and the government people who are involved in this, and the lawyers.

All of these people must be prosecuted with a forensic accounting and (unintelligible) land and their assets be taken, frozen for 300 years. We need 150-year back and 150 years going forward so that generations who have made illicit gains and profits cannot continue to reinvest. All of those profits need to be removed. All of the assets need to be taken.

And the people need to be put in prison cells for life. And you throw away the key. That's the beginning. That's just the beginning of what needs to happen here. All of this natural gas and all of its nonsense, and the companies who run it, must be shut down immediately, within a few months literally, with cell, solar and other renewable energies, which by the way are used…

Melanie Littlejohn: Thank you. Thank you.

Coordinator: And up next is (Kevin Corcoran). Your line is open. Please state your affiliation and ZIP code.

(Kevin Corcoran): Yes. I'm with the Central Jersey Safe Energy Coalition. And my ZIP code is 08540.

National Grid's long-term plan and the reason for the NESE project is that it is assumed that there's going to be an increasing demand for natural gas,

particularly for design days and the growth that National Grid expects for those in the future.

Just three comments. One, recent, let's say, increase in temperatures during the winter, such as what we just experienced this past winter, seemed to point in the opposite direction. Secondly, and I'm speaking to this in New York City, born and raised, I think that the COVID-19 problems we're going through right now will cause people to reconsider whether they want to live in dense population areas. Other areas have lower housing costs, lower rent, lower taxes. And I think people will start to move in that direction, and away from Brooklyn, Queens and Long Island.

Lastly, as corporations get more and more accustomed to working virtually, I think their people will also, and the corporations, will allow people to further and further afield in terms of where they're working from. And again, that would seem to point towards declining demand in Brooklyn, Queens and Long Island. Thank you.

Melanie Littlejohn: Thank you, (Kevin).

Coordinator: And next is (Joyce Cohen). Your line is open. Please state your affiliation and your ZIP code.

(Joyce Cohen): My affiliation is I'm a concerned citizen. My ZIP code is 10512. I wanted to start by also agreeing with those who have said that this platform for commentary is really inappropriate at this point with everything that's going on, particularly in New York. Additionally, it should be an addition to any public, real public platform where people could - where you could see the people's faces and deal with them as people, not as virtual voices.

Having said that, I wanted to indicate that I thought that all of the people that will be involved with the harming that this pipeline, the NESE pipeline could do by the Rockaways, I grew up enjoying the Hudson River and swimming and fishing, and loving the water, and living in the water every day in the summer. And I don't think these people deserve any less. They're taxpayers, they've been through a lot already.

And I don't see that it's fair or right to be putting them through the possibility of destroying their plant life, the water, and also interfere with the fishing, and just really the love of nature that this would be doing, especially also with all of the things that people have said of the dangers of fracking and what it does to the water by potentially poisoning it if there are leaks, essentially disrupting everything in the sediment that could be just staying there. There's just no way that it could be better for the people who live there than it is now.

I also wanted to say, I think it's a false choice as well. I don't see why we can't try to get electric costs down instead of saying everything is so expensive that we have to do, you know, X is so expensive so we have to do Y. Why don't we deal with X and try to get some regulations on the cost of electricity, besides other renewables?

And then lastly, I wanted to thank the activists who spend a lot of their free time living and loving these causes. And I really am very happy to be - to know that people do that, that they walk the walk and talk the talk. So, again, thank you for listening to my comments.

Melanie Littlejohn: Thank you, (Joyce).

Coordinator: Thank you. And there are no other callers at this time. As a reminder, it is star 1 if you would like to have your line open.

Melanie Littlejohn: We will hold and wait for the next caller.

Coordinator: I do have the next caller. That is (Herb Mens). Your line is open. And please state your affiliation and your ZIP code.

(Herb Mens): I'm a customer, 10301. I wanted to pick up on what has been mentioned already by other callers.

This is a disgraced utility. National Grid should not be in charge of New York's energy future. I'm shocked that the governor and other people are even allowing this nonsense to go forward. Basically, National Grid cares about profits. They want more fracked gas infrastructure to perpetuate an outdated business model, not because it needs the gas. So, you know, it's a multi-billion-dollar monopoly, basically, run by a small group of gas and oil criminals who are destroying our planet and killing our children.

And these people need to be treated as what they are. They're murderers. They're interested in killing…

Melanie Littlejohn: (Herb)…

(Herb Mens): …large numbers of people.

Melanie Littlejohn: (Herb), thank you for your comments.

Coordinator: And there are no other callers at this time. Once again, it is star 1 if you would like to have your line open.

And I do have a couple more callers. Up next is (Laura Shindell). Your line is open. Please state your affiliation and ZIP code.

(Laura Shindell): Hi there. 10019, Food and Water Action. If (Herb) is still on the call, I'd like to cede my time to him.

(Herb Mens): Hello.

Coordinator: And (Herb Mens), your line is open. Please state your affiliation and ZIP code.

(Herb Mens): Thank you. My name is (Herb Mens). I'm a customer, 10301. I would appreciate you not cutting me off in the middle of a sentence again please.

We pay for these pipelines so that the few investors and owners can make their trillions of dollars in profit killing our children and our women and our men, and destroying the atmosphere forever. They can't be returned. So we need to shut these people down forever.

And that means that we need to take the governor and the various legislators who allow for this nonsense to happen off - out of their seats, by any means possible, preferably offering them an opportunity to stop this within the next few weeks. I mean, we're dealing with a pandemic, so, you know, we have to understand that there are only so many hands-on deck right now.

But this company can be shut down and the criminals brought to trial and the infrastructure replaced within a matter of months, literally. I mean, Denmark, Sweden, countries in Central and South America, you know, entire cities and countries run on renewable energy. There's no need to have a Natural Grid

even in existence, a National Grid in existence. And it's very easy and much cheaper to run on renewable, solar and other types of renewable.

And the people who work for National Grid would just simply be retrained to do the same jobs and have many more jobs there. I mean, it's an incredible opportunity for jobs. So, people aren't going to be suffering. But the criminals who are killing us and making all the profit and paying people nickels and dimes for their work should suffer for what they're doing.

National Grid is a business model based on the profit of a few and the destruction of the many. This is a crime that we're dealing with here, a national and international crime, because billions of people are affected by the toxins that National Grid and other frackers create. They're poisoning the water in our - underneath our feet.

And, you know, when you go down 2500 feet, you drill down and you bring up all those toxins and all that radiation, which belongs down there, it doesn't belong at the 250 to 25-foot mark where our water is, and we're sitting on the Delaware Basin, one-fifth of the world's freshwater reserves, so we're poisoning the water for 8 billion people plus 100 million generations to come, so that the few criminals can make their trillions of dollars. These are greedy, sociopathic, Nazi criminals who wanted - and you have to be a sociopathic, Nazi criminal to want to poison the border and kill children.

I mean, if you're really that desperate to make a buck, go get yourself a job. National Grid makes its money off of us. And we need to shut that down immediately.

Cuomo and the rest of the people who are allowing this to happen should be impeached. And that's it.

Coordinator: And up next is (Jackie Weisberg). Your line is open. Please state your affiliation and ZIP code.

(Jackie Weisberg): (Jackie Weisberg), 350 Brooklyn, 11201.

We are in the epicenter of the pandemic, and we are in the midst of this climate crisis. And the two emergencies are in fact quite similar. Both had their roots in the world's current economic model, that of the pursuit of infinite growth at the expense of the environment on which our survival depend. And both are deadly and disruptive.

Harvesting natural resources of the planet, which is fossil fuel, and plugging them into an industrial cycle (unintelligible) various consumables like plastics and energy produces a ton of waste. And this process depletes the natural ability of the environment to balance itself and disrupt ecological cycles by producing carbon dioxide from burned fossils (unintelligible) methane.

This in turn is leading to changes in the climate of our planet. The same plastic is also responsible for the coronavirus and other outbreaks. So the need for more natural resources has forced humans to encroach our various natural habitats and expose themselves to yet unknown (pathogens).

Air pollution, largely from fossil fuel and (unintelligible) sources that also drive climate change is a key factor that worsens any given virus's impact on human health. A study of the 2003 outbreak of the severe acute respiratory syndrome, also known as SARS, in Asia, found that people exposed to the highest level of air pollution were twice likely to die from the disease as those who were not.

To combat climate change, we need to drastically decrease our greenhouse gas emissions from fossil fuels like coal, oil and natural gas. Generating electricity from low-carbon energy sources like wind and solar decreases harmful air pollutants such as nitrogen oxide, sulfur dioxide, methane gas, and carbon dioxide, that lead to more (unintelligible) and stroke, as well as obesity, diabetes, and premature death, that puts further strains on our health system.

Reducing air pollution also helps to keep our lungs healthy, which can protect us from respiratory infections like this current coronavirus.

Melanie Littlejohn: Thank you. Thank you.

Coordinator: And up next is (Brenda). Your line is open. Please state your affiliation and your ZIP code.

(Brenda): (Brenda), citizen, 10706. Energy. We all need it. We all need energy. Okay? The source of the energy is where we're at now and we cannot be going towards these fossil fuels. National Grid is not being straightforward with the (NYCHA) housing residents, the fellow that called in earlier. They are not, National Grid is not obligated to provide gas to (NYCHA) from the Williams pipeline. They're not - they're a private company.

And they can take this gap and they can decide, when the project is done, after New York residents have paid for this, they can decide to sell that, to (unintelligible) and sell it somewhere else, because they're a private company. And if that is what they want to do to make more money off of this project, after having residents pay for the project, they have every right to do that, because they're a private company.

There is not a question of if there'll be a Hurricane Sandy 2, it's when. When is it going to happen? And those pipes will be right off of Coney Island, right by the (NYCHA) houses, right by Breezy Point, right by (Ed) in Breezy Point, right by Jamaica Bay Wildlife Refuge Center, in New Jersey, in these very delicate waters.

And they will leak and they will crack, and they will break. And there is no containing it. Right? We learned that with (unintelligible). There's no containing it. There's no, like, mitigating. They like to use this word mitigating. There's no mitigating it. Once it's in the water, it's in the water. And that's methane and all kinds of stuff.

We need union jobs. I'm a union member. We need them in renewable energy. We need good jobs. The way in New York is something that is not discussed by National Grid. And you cannot continue to withhold this very key information. Pennsylvania is doing the fracking because of our laws in 2014 that were passed by very concerned citizens and politicians to ban fracking in New York State. So, Pennsylvania fracked and New York State, many people don't know this, is accepting a lot of the wastes. That was the deal.

The waste is very toxic. Very toxic to the land, and gets into the groundwater. We had this written about in 1963, let me just…

Coordinator: I do have another caller that is (Laura Miller). Your line is open. Please state your affiliation and your ZIP code.

(Laura Miller): Hi. My name is (Laura Miller), ZIP code 11211. I'm absolutely opposed to the pipeline. I can't believe we're in a time of pandemic, when people are frightened of the future, that National Grid is going to make us even more

frightened by - with the technology that is only going to do harm, and it will only do benefit to people - a small number of people who stand to make money. And I'd like to see the rest of my time either to the previous caller, so she can continue, or to (Herb). Thank you.

Melanie Littlejohn: Thank you, (Laura).

Coordinator: And I do have (Herb Mens) back. Your line is open. And please state your affiliation and ZIP code.

(Herb Mens): I'm a customer, 10301. I think that, since there are a lot of us who are concerned, probably most of us, and the information needs to be brought out to millions more, I think that en masse we should contact Food and Water Watch and other groups that have the ability to put together an impeachment of the people that were involved, the politicians who were involved in allowing National Grid and other hydro-fractors to function. And I think that we need to insist upon immediate arrests, prosecutions of the people who are running National Grid and the other hydrofracking companies and oil companies.

The poison in the area here all over, there isn't a day, including tonight, that I don't smell petrochemicals pouring through my window here, I'm about 20 minutes from Bay Own. So the National Grid oil hydrofracking groups of criminals must be stopped immediately, because they're poisoning the air and the water and the land and these people are also investing, these oligarchs are investing in burning down all the trees and the forests around the world and receding them with (NGL), you know, seeds, genetically modified seeds, so that essentially they're poisoning enough of the land and enough of the water so that they're going to enslave us, because nobody can live without freshwater for more than three days, and you bring in the 5G and which is sort

of like throwing tinfoil in a microwave, and they'll be able to fry us alive from the inside from the satellites.

So this isn't just about one area of poisoning of the world. This is about a total takeover, which is what you're seeing with the current administration, this sociopath named Donald Trump. We are looking at a carefully engineered global food and water shortage that is - and people are essentially worldwide under martial law right now. And this idiot calls himself a wartime President because he's looking to start a war or continue the war, his family are Nazi…

Melanie Littlejohn: Thank you, (Ed). Thank you.

Coordinator: And we do have another caller that is (Eileen Berry). Your line is open. Please state your affiliation and ZIP code.

(Eileen Berry), your line is open.

(Eileen Berry): Hi. 11554, I'm a citizen. And all the information that's been brought out that we should have wind turbines or solar panels, is anybody that's recommending all this realize where they come from and how they're made and how long they last. Their life expectancy is 20 to 30 years. They have to be rebuilt. When they're being rebuilt, what is used to build them and where are they being built? Most of them are made in China. Hello, people, you're so worried about having these conversations during this pandemic.

Well, grow up, at least you're focusing on something else instead of yourselves. And to worry about this man talking about Trump's derangement syndrome. Give me a break. Look at where these wind turbines are being made and the solar panels are being made, and the life expectancy of them. And when there's no sun or there's no wind, what do you have as backup?

Now there's a lot of people out here that have relied on natural gas for years without a problem. I've been looking for it for 17 years in my house trying to get hooked up and I cannot. There's a lot of people in New York who want it. Just lift these ridiculous regulations and let us have the pipeline that's safer than any of this other stuff that you're talking about. Thank you.

Melanie Littlejohn: Thank you, (Eileen).

Coordinator: And I do have another caller, that's (Herb Mens). Your line is open. Please state your affiliation and your ZIP code.

(Herb Mens): Customer, 10301. To the lady who just addressed me, with all due respect, there's a tremendous information gap, just all you need to do is look at what's happening to people who live in Pennsylvania which is, you know, about an hour from us here. And they're suffering. I mean, this is not - gas is not the opportunity that you think it is. It's shutting out jobs. It's closing the economy. It's poisoning people, of all ages. And it's destroying the earth.

But the - we don't have to - I agree with you, we don't have to buy from China or anywhere else. We can make them right here. And that's exactly what the new green deal represents. It represents a return to manufacturing here in the United States. We are the world's leader. A lot of that stuff, the technology came from us. And yes, we have to be careful how it's created, how these various means are created.

But geothermal has been around for thousands of years, and it's been very effective. So that's a renewable. Using the sun is a renewable. I mean, the danger of a broken pipe or of a bombtrain is - you can't compare it. I mean, it's forever. Whereas, if you don't have those kinds of issues, you could still

heat your house and live very well on storing the sun - on storing the energy from the sun. People do it all the time. Entire nations run on this.

So there are only a few people who are making money and the rest of us are paying for it. It's completely crazy.

Melanie Littlejohn: Thank you, (Herb). Thank you.

Coordinator: And up next is (Laura Shindell). Your line is open. Please state your affiliation and ZIP code.

(Laura Shindell): (Laura Shindell), ZIP code 10019, Food and Water Action. A previous caller just reminded me to mention that, you know, I think we all expect that we'll be in an economic rut following this health crisis. And green new deal guarantees green jobs and puts people to work. To build renewable energy will be a fantastic way to dig us out of the economic rut we anticipate to be in. And we should put people to work, building the renewable options that our climate and our earth so desperately need. (Previous) comment, that's all.

Melanie Littlejohn: Thank you, (Laura).

Coordinator: And up next is (Herb Mens). Your line is open. Please state your affiliation and your ZIP code.

(Herb Mens): Just a customer, 10301. National Grid's new report showing the need for more infrastructure like the Williams NESE pipeline is full of omissions, skewed data, and essentially propaganda. The entire 116-page report is based on grossly inflated gas demand projects that do not match up with the figures, provided by the Energy Information Administration. The National Grid's

demand projects also fail to reflect the utility's own historical trends which show gas used to be declining.

The report downplays the contributions that renewable heat pumps, which double as air-conditioners, demand response technology, efficiency, and other measures could make in reducing demand, while failing to mention whether it could offset pipeline demand through innovations in the power sector, which accounts for 42% of National Grid's gas sales on Long Island, and 20% in New York City alone.

So the report doesn't mention that rate payers would be bearing the costs of these projects. That means that we're spending our hard-earned money to make a bunch of criminals trillions of dollars more money than they already have. Why they need more money when everybody is starving and out of work, I don't know. But why they need to be paid to poison us and destroy our future and kill our children, I really don't - I don't understand.

Maybe they're spending a lot of their money paying off all of the criminals in the government and so they really need us to pay them. But National Grid's report includes pipeline alternatives such as offshore and LGN - LNGs which are these horrific, you know, whether they call these…

Melanie Littlejohn: Thank you, (Herb). Thank you. Thank you.

Coordinator: Thank you. And the next caller is (Tanya Jones). Your line is open. Please state your affiliation and ZIP code.

(Tanya Jones): Hi. My name is (Tanya Jones). My ZIP code is 11421. And I am from HeartShare Human Services.

I'm listening to the callers, some very upset for the pipeline, some just for getting gas to their neighborhood. I think, I'm not a scientist, I'm not an environmentalist. I've been dealing with low-income population and low to moderate income populations over the last 23 years.

I can tell you right now that fossil fuels, oil accounts, they're not affordable. People cannot afford it. And I have (unintelligible) offices and community centers and senior centers and just in my office we have people crying because they can't afford their electric bill, they can't afford an oil bill. They're looking to convert to gas.

Looking at all of these new constructions just popping up everywhere. All of these buildings, all of these new houses, people trying to move to where it's affordable. They can't afford fossil fuels. They can't afford to have electric accounts. They need gas.

I know that there's a lot of reports coming out saying, you know, National Grid can - has, you know, over-exceeded or overestimated the amount of accounts they're going to have and all of that. We really don't know that, because of all of the new construction and all of the people moving in to New York, moving out of New York, having, you know, families, losing their jobs, possibly needing something more affordable.

You just don't know. So you can't say, oh my God, you know, they are overestimating what they may need, when there are so many people who have oil now who need to convert. There are people who are going to be moving into these new buildings and they're going to need a gas account.

You just have to look at all sides. Thank you.

Melanie Littlejohn: Thank you, (Tanya).

Coordinator: Thank you. And up next is (Eileen Berry). Your line is open. Please state your affiliation and ZIP code.

(Eileen Berry): Hi. I'm a citizen, 11554. I heard a couple of people talk about the green new deal, which is wind turbines and the solar panels. I don't know if this realize that the state-of-the-art wind turbine blades are made of carbon fiber, which consists of layers of plastic and plastic resin, both of which are derived from oil and natural gas feedstocks. Many components of the turbines and solar panels are manufactured where? Not in the United States.

Well, you did say, well, maybe it will bring the jobs back here in the United States. But right now they're manufactured in China, which then they have to be shipped here, on ships, which burn heavy fuel, heavy diesel fuel and oil. The components of the wind turbines and solar panels are transported from coast to coast across the United States in, what? Oh, that's right, not solar-run trucks but diesel-run trucks, across our roads, which is just killing our environment even more.

This is a significant waste, this green new deal. Natural gas is the way to go. You had said that all these solar stuff has been around for years and years and years. You don't see us living that way and we haven't lived that way. Why? Because it wasn't beneficial. And now, 20 to 30 years, when these solar panels and wind turbines are no good anywhere, what are we going to do with all those junk steel that was made of fossil fuel? What are we going to do with this? You have a plan for that in your green new deal? Give me a break.

We've been here begging for this natural gas to come here. Just let it go. Thank you.

Melanie Littlejohn: Thank you, (Eileen).

Coordinator: Thank you. And the next caller is (Herb Mens). Your line is open. Please state your affiliation and your ZIP code.

(Herb Mens): I'm a customer, 10301. I agree with part of what that lady had to say. We need to set up across the country, and starting right here in the city and throughout the area here, the electric refueling stations for automobiles and for trucks. And I know that they're rolling these out now. But we need to expedite that. That's got to be part of the infrastructure that will supply millions and millions and millions of jobs.

And the plan has to include what we do with this, you know, with these various items such as wind turbines and their blades at a point when they are no longer serviceable. So these are very, very good points. But the immediate need to shut down and replace the gas and the oil and other kinds of fossil fuels that will destroy - that are destroying our environment as we speak, and replace them immediately with renewable processes that give people an inexpensive alternative and a far more affordable opportunity that should be paid for by the fossil fuel companies and by the oil companies, not by us, the taxpayers. Because they've made trillions of dollars on us for the last 140 years or so. So let them give back for a change, and let's turn this over.

The lady talked about - there was one lady who talked about people crying because they can't afford gas or oil, hydrofracked or not. In fact, if we turned it back around, those individuals should be making money on the sun and then giving that back into the network. So that's part of a new green deal concept.

None of this stuff is new. I mean, these suggestions and this technology has been around for, I think, some of it's been around for thousands of years, but more recently, you know, for 50 years or so…

Melanie Littlejohn: Thank you, (Herb). Thank you.

Coordinator: And as a reminder, it is star 1 if you would like to have your line open.

I do have a caller. (Brenda), your line is open. Please state your affiliation and your ZIP code.

(Brenda): (Brenda), citizen, 10706. I'd like to pick up on the very important point that (Tanya) was making about affordability. I'd like to make clear that there is a possibility of having public utilities versus private utilities. This is something that (Jackie) was touching on earlier, in terms of the way our entire system is structured.

Everyone needs heat, everyone needs energy. And I understand what (Tanya) needs to be heard, because it does affect very badly poor people, lower-income people, to have to use private companies. But this, National Grid is a privately owned company, with their private investors. There is a possibility having public utility companies where your source would be much more regulated and the amount of - and the amount of money that you will be paying, the customers will be paying per month will be much more regulated.

During the Bloomberg administration, we had a chance to move, he got us off of oil number 6, I believe, and put everybody onto oil number 2, which is a cleaner dirty oil. And I think I got that in the right order. And he really pushed and gave a lot of tax breaks to private companies to turn the entire city of New York over to natural gas.

This is a disaster because we could have put in a lot of geothermal in Manhattan in the last 10 years. St. Patrick's Cathedral is heated and cooled with geothermal. It's a perfect location for geothermal, its bedrock.

Okay, (Eileen Berry) keeps bringing up solar and wind and where things are produced. Not everything is solar and wind. Not all energy has to come from those two sources, or oil, or - and you keep referring to it as natural gas as though it is pure and clean. It is a label, just like natural eggs are labeled at the grocery store. It's kind of meaningless actually. And you have to really think about the words that have been put into the general public's vernacular. It appears that this is good, clean energy. It is not renewable and it is not clean.

And at this point, let me just finish, at this point, we are using fracked gas. This is not coming from a very big reserve. This is being fracked. It is dirtying the water of Pennsylvania. And in good conscience, nobody in New York should feel good about the fact that we're (filthying) another state. By the way, we're taking the waste in western New York.

Melanie Littlejohn: We have time for one more caller.

Coordinator: Thank you. And that comes from (David Vassar). Your line is open. Please state your affiliation and your ZIP code.

(David Vassar): Hi. I'm with Food and Water Action. My ZIP is 10027.

Yes, I just, you know, really want to say that a green new deal is what we all need now more than ever, especially in this time of a public pandemic which we're all, all of us are frightened about, understandably.

So, I mean, wind and solar, as well as geothermal, and even hydro, these are not called renewable for no reason. I mean, they are there and will be in place for, you know, for basically the life of the planet. And so, investing in these is a real investment. The investment will pay for itself over time and will show returns, not only in terms of the energy that's produced, and as well I would say far healthier environment, but economically as well.

I mean, gas and oil is just such a short-term commodity and it's basically diminishing returns right now. We're having to frack and destroy, you know, the earth's strata deeper and deeper, which is bringing up poisoned water and reservoirs, as well as destruction, literal destruction to the communities affected. And yes, Pennsylvanians should not have to pay so dearly for New York's energy needs. I mean, fracked gas is neither environmentally nor morally sustainable.

The word natural is a real misnomer. I mean it's natural only in the sense that it's a poison. It's natural poison.

So, solar panels, yes, I mean, America has the knowhow. We should be making those here. I mean, we have got what it takes, and that's true for turbines, that's true for geothermal apparatus. Solar panels now does better economically than oil or gas, just for wattage produced per unit.

So, and yes, we have to ensure that all communities benefit from the green new deal and renewable energy. I mean, we can't have some folks going cold in the winter because they lack the infrastructure. But fracked gas does far more harm than good. It brings some heat, you know, in the short term. But I mean, the poison of our water supply, our air through methane releases, and the disruption of our - the strata of the earth we live on is completely…

Melanie Littlejohn: Thank you, (David). Thank you. Thank you.

And again, we appreciate all comments. Thank you. That concludes tonight's comment period. Thank you for joining us tonight. The feedback and the questions collected here are certainly a valuable part of the public input process. National Grid plans to issue a supplemental report in late April that will reflect the public input we received, all to enable an agreed upon long-term solution or solutions with New York State by June 2020. This timeline will enable the solution or solutions to be in place and in operation by the winter of '21/'22.

As a reminder, if you have further questions or input, please provide it directly to the New York Public Service Commission. We have instructions for doing so on our report Web site.

That concludes tonight's call. Thank you and be well.

Coordinator: Thank you. That does conclude today's conference. We appreciate your participation. You may disconnect at this time. Speakers, stand by for your post-conference.

END